# Systemic Therapy for Melanoma: ASCO Guideline Update

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## ABSTRACT

**PURPOSE** To provide guidance to clinicians regarding the use of systemic therapy for melanoma.

**METHODS** American Society of Clinical Oncology convened an Expert Panel and conducted an updated systematic review of the literature.

**RESULTS** The updated review identified 21 additional randomized trials.

**UPDATED RECOMMENDATIONS** Neoadjuvant pembrolizumab was newly recommended for patients with resectable stage IIIB to IV cutaneous melanoma. For patients with resected cutaneous melanoma, adjuvant nivolumab or pembrolizumab was newly recommended for stage IIB-C disease and adjuvant nivolumab plus ipilimumab was added as a potential option for stage IV disease. For patients with unresectable or metastatic cutaneous melanoma, nivolumab plus relatlimab was added as a potential option regardless of *BRAF* mutation status and nivolumab plus ipilimumab followed by nivolumab was preferred over BRAF/MEK inhibitor therapy. Talimogene laherparepvec is no longer recommended as an option for patients with *BRAF* wild-type disease who have progressed on anti–PD-1 therapy. Ipilimumab- and ipilimumab-containing regimens are no longer recommended for patients with *BRAF*-mutated disease after progression on other therapies.

This full update incorporates the new recommendations for uveal melanoma published in the 2022 Rapid Recommendation Update.

Additional information is available at www.asco.org/melanoma-guidelines.

ACCOMPANYING CONTENT

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Appendix

Data Supplement

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## INTRODUCTION

Systemic treatment for melanoma has changed rapidly since the introduction of ipilimumab in 2011.[1] In the past decade, 12 new drugs have been approved for unresectable melanoma, along with four new approvals in the adjuvant setting and increasing research in neoadjuvant therapy. Overall survival for melanoma has improved substantially,[2] with dramatic increases in long-term survival observed from administration of these new therapies. Multiple therapeutic options and longer survival in melanoma have led to increased consultation hours and also to increased patient and societal costs. In addition, the clinical burden associated with the treatment of melanoma has been increasing because of rising incidence in most countries worldwide. In some countries, melanoma incidence has grown by 3% annually.[3] In 2023, 97,620 new diagnoses and 7,990 deaths are projected in the United States.[4] Worldwide, the International Agency for Research on Cancer estimated that in 2020, there were 324,635 new diagnoses and 57,043 deaths from melanoma.[5] With both melanoma incidence and treatment cost rising, rational evidence-based selection of appropriate therapy is essential.[6]

Because of the morbidity and mortality associated with melanoma and the rapid changes in therapeutic options, ASCO developed a guideline on systemic therapy for melanoma in 2018 that made recommendations on adjuvant and metastatic therapies. However, since that time, further randomized trials have been published addressing newer agents, neoadjuvant therapy, therapy for patients with earlier-stage disease, and uveal melanoma. Given this large quantity of new data and the

## THE BOTTOM LINE

**Systemic Therapy for Melanoma: ASCO Guideline Update**

**Guideline Question**

What systemic therapy options have clinical benefit in adult patients with melanoma?

**Target Population**

Adult patients with melanoma (cutaneous and noncutaneous).

**Target Audience**

Clinicians who treat patients with melanoma.

**Methods**

An Expert Panel was convened to develop clinical practice guideline recommendations on the basis of a systematic review of the medical literature.

**Recommendations**

NOTE: All references to stage in these recommendations refer to stage determined by the eighth edition American Joint Committee on Cancer criteria[7] unless otherwise noted.
These recommendations are summarized in the algorithm depicted in Figure 1.

*Recommendation 1.1*

Neoadjuvant pembrolizumab (maximum of three courses of 200 mg once every 3 weeks) followed by resection and adjuvant pembrolizumab (maximum of 15 courses of 200 mg once every 3 weeks) should be offered to patients with clinical and resectable stage IIIB-IV cutaneous melanoma (Type: Evidence based, benefits outweigh the harms; Evidence quality: Moderate; Strength of recommendation: Strong).

*Recommendation 1.2*

Patients with clinical and resectable stage IIIB-IV cutaneous melanoma should be offered or referred for enrollment in clinical trials of neoadjuvant therapy where possible (Type: Informal consensus; Evidence quality: Not applicable; Strength of recommendation: Strong).

*Recommendation 2.1.1*

Adjuvant pembrolizumab or nivolumab should be offered to patients with resected stage IIB or IIC melanoma. See Table 1 for reasonable doses and schedules (Type: Evidence based; Evidence quality: Moderate; Strength of recommendation: Strong).

**Qualifying statement.** The recommendation in favor of nivolumab is based at the time of writing on the conference abstract publication of CheckMate 76K[13] and under the assumption that full publication of those data will not contradict the abstract. See the Clinical Interpretation section for further details.

*Recommendation 2.1.2*

Adjuvant therapy should not be offered to patients with resected stage IIA melanoma outside of enrollment in a clinical trial (Type: Informal consensus; Evidence quality: Not applicable; Strength of Recommendation: Strong).

*Recommendation 2.2*

For patients with resected stage IIIA-D disease that is *BRAF* wild-type, the following adjuvant therapy options should be offered (in no particular order): nivolumab × 52 weeks **OR** pembrolizumab × 52 weeks. Ipilimumab and high-dose interferon are not recommended for routine use in adjuvant therapy. See the Clinical Interpretation section on shared decision making with patients between these options. See Table 1 for recommended dosing and scheduling details (Type: Evidence based, benefits outweigh the harms; Evidence quality: High; Strength of recommendation: Strong).

**Qualifying statements.** Patients with stage IIIA disease with microscopic sentinel nodal metastasis < 1 mm in diameter were not included in the randomized trials that studied efficacy of immune checkpoint inhibitors as adjuvant therapy for melanoma. Both nivolumab and pembrolizumab are US Food and Drug Administration (FDA)–approved adjuvant treatments for patients with melanoma with lymph node involvement who have undergone complete disease resection. Patients with stage IIIA disease with <1 mm involvement in the sentinel lymph node have a relatively better prognosis and lower risk of relapse. Therefore, treatment should be individualized after discussing risk-benefit quotient with these patients.

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**THE BOTTOM LINE (CONTINUED)**

*Recommendation 2.3*

For patients with resected stage IIIA-D disease that is *BRAF* mutant (V600E/K*), the following adjuvant therapy options should be offered (in no particular order): nivolumab × 52 weeks OR pembrolizumab × 52 weeks OR dabrafenib plus trametinib × 52 weeks. See the Clinical Interpretation section on shared decision making with patients between these options. See Table 2 for reasonable dosing and scheduling details (Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

**Qualifying statements.** Patients with stage IIIA disease with microscopic sentinel nodal metastasis <1 mm diameter were not included in the randomized trials that studied efficacy of immune checkpoint inhibitors and BRAF-targeted agents as adjuvant therapy for melanoma. Nivolumab, pembrolizumab, and dabrafenib plus trametinib are FDA-approved adjuvant treatments for patients with melanoma with lymph node involvement who have undergone complete resection of their disease. Patients with stage IIIA disease with <1 mm involvement in the sentinel lymph node usually have a good prognosis and low risk of relapse. Therefore, treatment should be individualized after discussing risk-benefit quotient with these patients.

*Recommendation 2.4*

No recommendation can be made for or against adjuvant BRAF/MEK inhibitor therapy in patients with resected stage III/IV melanoma with *BRAF* mutations other than V600E/K (Type: No Recommendation; Evidence quality: NA; Strength of recommendation: NA).

*Recommendation 2.5.1*

Patients with resected stage IV melanoma should be offered adjuvant therapy (Type: Informal consensus; Evidence quality: Moderate; Strength of recommendation: Strong).

*Recommendation 2.5.2*

Reasonable options for therapy are nivolumab alone (Evidence quality: Moderate), nivolumab plus ipilimumab followed by nivolumab (Evidence quality: Low), pembrolizumab alone (Evidence quality: Low), and dabrafenib plus trametinib (in patients with *BRAF* V600E/K disease; Evidence quality: Low). See the Clinical Interpretation section on shared decision making with patients between these options (Type: Informal consensus; Evidence quality: varies by option; Strength of recommendation: Weak).

*Recommendation 3.1*

For patients with *BRAF* wild-type, unresectable and/or metastatic cutaneous melanoma, the following treatment options should be offered (in no particular order): nivolumab plus ipilimumab followed by nivolumab OR nivolumab plus relatlimab OR nivolumab OR pembrolizumab. See Table 2 for recommended dosing and scheduling details. See the Clinical Interpretation section on shared decision making with patients between these options (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

*Recommendation 3.2.1*

For patients with *BRAF* mutant (V600) unresectable and/or metastatic cutaneous melanoma, one of the following treatment options should be offered as first-line therapy: nivolumab plus ipilimumab followed by nivolumab OR nivolumab plus relatlimab OR nivolumab OR pembrolizumab OR dabrafenib plus trametinib OR encorafenib plus binimetinib OR vemurafenib plus cobimetinib. See Table 2 for recommended dosing and scheduling details. See the Clinical Interpretation section on shared decision making with patients between these options (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

*Recommendation 3.2.2*

Combination therapy with nivolumab plus ipilimumab is preferred as first-line therapy over BRAF/MEK inhibitor combination therapy. See the Clinical Interpretation section on shared decision making with patients between these options (Type: Evidence based; Evidence quality: High for dabrafenib plus trametinib, Low for other BRAF/MEK inhibitors; Strength of recommendation: Strong).

*Recommendation 3.3*

After progression on anti–PD-1–based therapy, patients with unresectable and/or metastatic *BRAF* wild-type cutaneous melanoma may be offered ipilimumab or ipilimumab containing regimens. Patients should be offered or

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## THE BOTTOM LINE (CONTINUED)

referred for enrollment in clinical trials where possible (Type: Informal consensus; Evidence quality: Low; Strength of recommendation: Weak).

*Recommendation 3.4*

After progression on first-line anti–PD-1–based therapy, patients with *BRAF* mutant (V600) unresectable and/or metastatic cutaneous melanoma may be offered combination BRAF/MEK inhibitor therapy as described in Recommendation 3.2.1. Similarly, those who have progressed after combination BRAF/MEK inhibitor therapy may be offered anti–PD-1–based therapy as described in Recommendation 3.2.1 (Type: Informal consensus; Evidence quality: Low; Strength of recommendation: Weak).

*Recommendation 3.5*

For patients with injectable (cutaneous or subcutaneous or nodal) unresectable lesions who are not eligible or do not desire the recommended systemic therapies, talimogene laherparepvec may be offered as primary therapy (Type: Evidence based, benefits outweigh harms; Evidence quality: Low; Strength of recommendation: Weak).

*Recommendation 4.1.1*

HLA-A*02:01-positive patients with metastatic uveal melanoma should be offered tebentafusp (20 μg once on day 1, 30 μg once on day 8, and 68 μg once weekly after that until progression; Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

*Recommendation 4.1.2*

For all patients with uveal melanoma other than those addressed by recommendation 4.1.1, no recommendation for or against any specific systemic therapy may be made at this time. Patients should be offered or referred for enrollment in clinical trials where possible (Type: No recommendation; Evidence quality: Very Low; Strength of recommendation: Not applicable).

*Recommendation 4.2*

In the absence of further data, the consensus of the Expert Panel is that patients with unresectable and/or metastatic mucosal melanoma may be offered therapy as described in recommendations 3.1 through 3.5. Patients should be offered or referred for enrollment in clinical trials where possible (Type: Informal consensus; Evidence quality: Very Low; Strength of recommendation: Weak).

*Recommendation 4.3*

No recommendation for or against any specific systemic therapy for patients with any other form of non-cutaneous melanoma may be made at this time. Patients should be offered or referred for enrollment in clinical trials where possible (Type: No recommendation; Evidence quality: Not applicable; Strength of recommendation: Not applicable).

**Additional Resources**

Definitions for the quality of the evidence and strength of recommendation ratings are available in Appendix Table A1 (online only). More information, including a supplement with additional evidence tables, slide sets, and clinical tools and resources, is available at www.asco.org/melanoma-guidelines. The Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the methods used to develop this guideline. Patient information is available at www.cancer.net

**ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care and that all patients should have the opportunity to participate.**

importance of these data to practice, ASCO determined that the 2018 guideline should receive a full update.

## GUIDELINE QUESTIONS

This guideline update addresses the same four clinical questions addressed in the 2018 guideline: (1) What neoadjuvant systemic therapy options, alone or in combination, have demonstrated clinical benefits in adults with cutaneous melanoma eligible for resection? (2) What adjuvant systemic therapy options, alone or in combination, have demonstrated clinical benefits in adults with resected (stage II, stage III, stage IV) cutaneous melanoma? (3) What systemic therapy options, alone or in combination, have demonstrated clinical benefits in adults with unresectable and/or metastatic cutaneous melanoma? (4) What systemic therapy options, alone or in combination, have demonstrated clinical benefits in adults with noncutaneous melanoma (stage II or greater)? All clinical questions also addressed the subquestion: Are there subpopulations of patients (ie, clinical features, biomarker status, specific type of melanoma) who benefit more or less from those options?

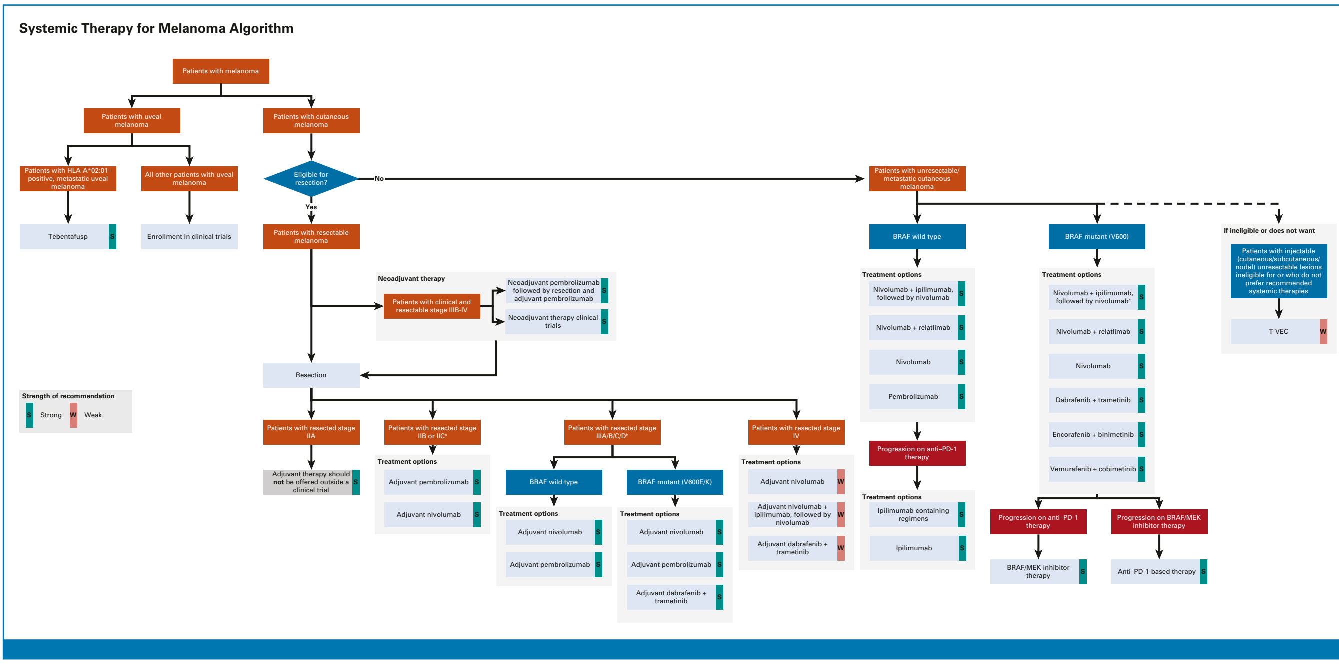


**FIG 1.** Algorithm for systemic therapy for melanoma. [a]While the consensus of the Panel was that a strong recommendation could be made for adjuvant nivolumab or pembrolizumab for these patients, the potential for serious morbidity, both acute and long term, because of immune-related toxicity must be considered and weighed against the lack of overall survival data in stage II disease. Treatment should be individualized after discussing risk-benefit quotient with these patients. [b]Patients with stage IIIA disease with microscopic sentinel nodal metastasis <1 mm in diameter were not included in the randomized trials that studied efficacy of immune checkpoint inhibitors as adjuvant therapy for melanoma. Both nivolumab and pembrolizumab are US Food and Drug Administration–approved adjuvant treatments for patients with melanoma with lymph node involvement who have undergone complete disease resection. Patients with stage IIIA disease with <1 mm involvement in the sentinel lymph node have a relatively better prognosis and lower risk of relapse. Therefore, treatment should be individualized after discussing risk-benefit quotient with these patients. [c]This option has been found to be superior to dabrafenib plus trametinib in a phase III trial. T-VEC, talimogene laherparepvec.

## METHODS

### Guideline Development Process

This systematic review-based guideline product was developed by a multidisciplinary Expert Panel of medical, radiation, surgical, and community oncologists as well as two patient representatives and an ASCO guidelines staff member with health research methodology expertise. The Expert Panel met via teleconference and webinar and corresponded through e-mail. On the basis of the consideration of the evidence, the authors were asked to contribute to the development of the guideline, provide critical review, and finalize the guideline recommendations. The guideline recommendations were sent for an open comment period of 2 weeks, allowing the public to review and comment on the

**TABLE 1.** Reasonable Dose and Schedule Details for Recommended Regimens—Neoadjuvant and Adjuvant Therapies

| Regimen (Recommendation No.) | Dosing Schedules (source)[a] |
|---|---|
| Neoadjuvant plus adjuvant pembrolizumab (1.1) | Preresection: pembrolizumab, maximum of 3 courses of 200 mg iv once every 3 weeks<br>Postresection: pembrolizumab, maximum of 15 courses of 200 mg iv once every 3 weeks (SWOG S1801[8]) |
| Pembrolizumab × 52 weeks (2.1.1, 2.2, 2.3, 2.5.2) | Pembrolizumab 200 mg iv once every 3 weeks for 52 weeks (EORTC 1325[9]) |
| Nivolumab × 52 weeks (2.1.1, 2.2, 2.3, 2.5.2) | Nivolumab 240 mg iv once every 2 weeks or once 480 mg iv once every 4 weeks for 52 weeks (CheckMate 238[10] and US FDA label) |
| Dabrafenib plus trametinib × 52 weeks (2.3, 2.5.2) | Dabrafenib 150 mg orally twice daily plus trametinib 2 mg orally once daily for 52 weeks (COMBI-AD[11]) |
| Nivolumab plus ipilimumab followed by nivolumab (2.5.2) | Nivolumab 1 mg/kg iv plus ipilimumab 3 mg/kg iv once every 3 weeks for four doses, followed by nivolumab 3mg/kg once every 2 weeks for 1 year or until disease recurrence, whichever comes first (IMMUNED trial[12]) |

NOTE: These were doses and schedules considered reasonable at the time of publication of this guideline. Additional doses and schedules might have been approved at the time of reading.
Abbreviations: EORTC, European Organisation for Research and Treatment of Cancer; FDA, US Food and Drug Administration; iv, intravenously.
[a]A particular regimen might have been used in many trials, and only one source is provided.

**TABLE 2.** Reasonable Dose and Schedule Details for Recommended Regimens—Unresectable and/or Metastatic Disease

| Regimen (Recommendation No.) | Dosing Schedules (source) |
|---|---|
| Nivolumab plus ipilimumab followed by nivolumab until disease progression (3.1, 3.2.1) | Nivolumab 1 mg/kg plus ipilimumab 3 mg/kg iv once every 3 weeks for 4 doses, followed by nivolumab 3 mg/kg iv once every 2 weeks (CheckMate 067[14])<br>Nivolumab 1 mg/kg plus ipilimumab 3 mg/kg iv once every 3 weeks for 4 doses, followed by nivolumab 240 mg iv once every 2 weeks or 480 mg every 4 weeks (US FDA prescribing information) |
| Nivolumab plus relatlimab (3.1, 3.2.1) | Relatlimab 160 mg and nivolumab 480 mg iv once every 4 weeks until progression (RELATIVITY-047[15]) |
| Nivolumab (3.1, 3.2.1) | Nivolumab 3 mg/kg iv once every 2 weeks (CheckMate 067[14])<br>Nivolumab 240 mg iv once every 2 weeks (US FDA–approved)<br>Nivolumab 480 mg iv once every 4 weeks (US FDA–approved) |
| Pembrolizumab (3.1, 3.2.1) | Pembrolizumab 10 mg/kg iv once every 2 weeks or once every 3 weeks (KEYNOTE 006[16])<br>Pembrolizumab 2 mg/kg iv once every 3 weeks (KEYNOTE 002[17])<br>Pembrolizumab 200 mg iv once every 3 weeks (US FDA–approved)<br>Pembrolizumab 400 mg iv once every 6 weeks (EU EMA–approved [on the basis of the study by Lala et al[18]]) |
| Vemurafenib plus cobimetinib (3.2.1) | Vemurafenib 960 mg orally twice daily without pause plus cobimetinib 60 mg orally once daily in a cycle of 21 days with 7 days off (CoBRIM[19]) |
| Dabrafenib plus trametinib (3.2.1) | Dabrafenib 150 mg orally twice daily plus trametinib 2 mg orally once daily (COMBI-v[20]) |
| Encorafenib plus binimetinib (3.2.1) | Encorafenib 450 mg orally once daily plus binimetinib 45 mg orally twice daily (COLUMBUS[21]) |

NOTE: These were doses and schedules considered reasonable at the time of publication of this guideline. Additional doses and schedules might have been approved at the time of reading.
Abbreviations: EMA, European Medicines Agency; EU, European Union; FDA, US Food and Drug Administration; iv, intravenously.



recommendations after submitting a confidentiality agreement. These comments were taken into consideration while finalizing the recommendations. Members of the Expert Panel were responsible for reviewing and approving the penultimate version of the guideline, which was then circulated for external review and submitted to the *Journal of Clinical Oncology* (*JCO*) for editorial review and consideration for publication. All ASCO guidelines are ultimately reviewed and approved by the Expert Panel and the ASCO Evidence-Based Medicine Committee (EBMC) before publication. All funding for the administration of the project was provided by ASCO.

The recommendations were developed by using several resources:

- A group systematic review published by Pasquali et al[22] (the Cochrane review) that included trials of systemic therapy for metastatic cutaneous melanoma published in 2016 or earlier.
- A systematic review conducted by ASCO staff for the 2020 guideline[23] of randomized clinical trials (RCTs) of adjuvant and neoadjuvant systemic therapies for cutaneous melanoma, systemic therapy for metastatic melanoma published after 2016, and systemic therapy for noncutaneous melanoma. The methods of that review are fully described in the original 2020 guideline publication.[23]
- An updated systematic review conducted by ASCO staff of RCTs published since the 2020 guideline, described here.

This full update also incorporates all data and alterations to recommendations from the 2022 Rapid Recommendation Update[24] regarding tebentafusp in uveal melanoma.

PubMed was searched on March 6, 2023, for studies published on June 1, 2019, or later to that date, and articles were selected for inclusion on the basis of the following criteria:

- Population: Patients with any form of melanoma
- Interventions and comparisons: Trials of systemic therapy versus other systemic therapy or observation or placebo.
- Fully published reports of RCTs with at least 50 patients per arm. The restriction of 50 patients per arm was not present in the original 2020 guideline criteria.

Meta-analyses of randomized trials were reviewed and are mentioned in the text when they provided additional information beyond what was available in the trials themselves. In addition, conference abstracts of trials for which no peer-reviewed publication was found are briefly summarized in the Data Supplement (Data Supplement 1, Table 6). These abstracts were used primarily to judge the likelihood of changes to the recommendations on the basis of future publications or were considered in the development of recommendations in a few cases noted in the literature review sections of those recommendations. A search of the Cochrane Central Registry of Controlled Trials conducted on May 12, 2022, supplemented this search and used the same criteria.

Articles were excluded from the systematic review if they were editorials, commentaries, letters, news articles, case reports, or narrative reviews. Non-English articles were obtained if they passed title and abstract review but could not be formally evaluated; their references were included if they were potentially relevant. The full details of the search strategy and search results are given in the Supplement. Randomized trial quality was assessed using methods on the basis of the Cochrane Risk of Bias tool[25] and elements of the

GRADE quality assessment and recommendations development process.[26] GRADE quality assessment labels (ie, high, moderate, low, very low) were assigned for each outcome relevant to recommendations considered for updating by the project methodologist in collaboration with the Expert Panel cochairs and reviewed by the full Expert Panel. In addition, a guideline implementability review was conducted. This review did not reveal any concerns, so no changes were made to the guideline text. Ratings for type and strength of the recommendation, and overall evidence quality are provided with each recommendation; see Appendix Table A1 for definitions.

The ASCO Expert Panel and guidelines staff will work with cochairs to keep abreast of any substantive updates to the guideline. On the basis of formal review of the emerging literature, ASCO will determine the need to update. The ASCO Guidelines Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the guideline update process. This is the most recent information as of the publication date.

### Guideline Disclaimer

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### Guideline and Conflicts of Interest

The Expert Panel was assembled in accordance with ASCO's Conflict of Interest Policy Implementation for Clinical Practice Guidelines ("Policy," found at https://www.asco.org/guideline-methodology). All members of the Expert Panel completed ASCO's disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting or advisory role; speaker's bureau; research funding; patents, royalties, other intellectual property; expert testimony; travel, accommodations, expenses; and other relationships. In accordance with the Policy, the majority of the members of the Expert Panel did not disclose any relationships constituting a conflict under the Policy.

## RESULTS

### Characteristics of Trials Identified in the Literature Search

The full details of systematic review for the 2020 guideline are described in the original guideline publication.[23] The characteristics of these trials and their data are briefly summarized here; refer to the 2020 guideline for full description. For the update, 21 RCTs of interest were identified.[8,12,15,27-44] The published studies included a wide range of systemic therapies in cutaneous and noncutaneous melanomas. Study design aspects related to individual study quality, strength of evidence, strength of recommendations, and risk of bias were assessed. Refer to the Supplement for more information. The identified evidence is discussed with detailed analysis and interpretation by the clinical question it addressed and presented with the recommendations. While network meta-analyses were not the primary focus of the systematic review, relevant analyses that were identified during the search are mentioned in the appropriate literature review section.

Additional publications for trials previously identified in the 2020 guideline are briefly described in the Data Supplement (Table 7). Additional publications for trials included in the 2018 Cochrane Review[22] are briefly described in the Data Supplement (Table 8). Unless otherwise mentioned in the text, these additional publications were not considered relevant to recommendation development. Trials that were identified in the search that have only been published to date in the form of conference abstracts are summarized in the Data Supplement (Table 6). These trials are mentioned within the text where they provide important context or suggest future potential treatment options but are not used as the basis of recommendations. Trial registrations where no peer-reviewed publication or conference abstract could be located are listed in the Data Supplement (Table 9). These trials are mentioned within the text when the data they would provide when published could impact the recommendations.

## RECOMMENDATIONS

All references to stage in these recommendations refer to stage determined by the eighth edition American Joint Committee on Cancer (AJCC) criteria[7] unless otherwise noted.

Since the publication of the 2020 guideline ASCO has updated its definitions for quality of evidence and strength of recommendation statements. These statements have been updated to the new definitions even where the text of the recommendation is unchanged.

Throughout this guideline, the phrase "nivolumab plus ipilimumab" refers to a schedule of several cycles of the combination followed by single-agent nivolumab for a longer period. The shorter phrase is used as a simplification, and if some other regimen is being discussed it is specifically mentioned. See Tables 1 and 2 for the actual dosages, cycles, and scheduling of the recommended therapies.

The recommendations are summarized in the algorithm depicted in Figure 1.

### Clinical Question 1

What neoadjuvant systemic therapy options, alone or in combination, have demonstrated clinical benefits in adults with cutaneous melanoma eligible for resection? Are there subpopulations of patients (ie, clinical features, biomarker status) who benefit more or less from those options?

*Summary of Updated Results for Question 1*

Two new trials were identified. The trial reported by Dummer et al[27] in 2021 compared talimogene laherparepvec (T-VEC) followed by surgery with upfront surgery in patients with resectable stage IIIB-IV melanoma with injectable lesions. This trial was considered at high risk of bias. The Southwest Oncology Group (SWOG) S1801 trial[8] compared neoadjuvant pembrolizumab followed by adjuvant pembrolizumab with adjuvant pembrolizumab in patients with resectable stage IIIB to IVC melanoma. It was considered to have some concerns for risk of bias. Both trials are detailed in the Data Supplement (Tables 1-5).

*Recommendation 1.1*

Neoadjuvant pembrolizumab (maximum of three courses of 200 mg once every 3 weeks) followed by resection and adjuvant pembrolizumab (maximum of 15 courses of 200 mg once every 3 weeks) should be offered to patients with clinical and resectable stage IIIB-IV cutaneous melanoma (Type: Evidence based, benefits outweigh the harms; Evidence quality: Moderate; Strength of recommendation: Strong).

*Recommendation 1.2*

Patients with clinical and resectable stage IIIB-IV cutaneous melanoma should be offered or referred for enrollment in clinical trials of neoadjuvant therapy where possible (Type: Informal consensus; Evidence quality: Not applicable; Strength of recommendation: Strong).

#### Literature review and analysis.

*Summary of the 2020 guideline.* Four randomized trials were identified: Amaria 2018A[45] compared neoadjuvant plus adjuvant dabrafenib plus trametinib versus contemporary standard of care; Amaria et al,[46] compared neoadjuvant ipilimumab plus nivolumab versus nivolumab, both followed by adjuvant nivolumab; Blank et al[47] compared neoadjuvant and adjuvant ipilimumab plus nivolumab with solely adjuvant ipilimumab plus nivolumab; and OpACIN-neo,[48] a follow-up to Blank 2018, compared two variant schedules of neoadjuvant ipilimumab plus nivolumab. None of these trials were considered sufficient support for recommendations in favor of neoadjuvant therapy. See the 2020 guideline publication and Data Supplement (Tables 2020-1 to 2020-5) for additional details.

*Updates for the 2023 guideline.* The Dummer et al[27] 2021 trial reported improved recurrence-free survival (RFS) hazard ratio (HR) 0.75 (95% CI, 0.58 to 0.96), event-free survival (EFS) HR 0.58 (95% CI, 0.44 to 0.78), and overall survival (OS) HR 0.49 (95% CI, 0.30 to 0.79), all in favor of T-VEC followed by surgery versus upfront surgery. 5.5% of patients had grade 3 or worse treatment-related adverse events (TRAEs) before surgery with T-VEC.

The SWOG S1801 trial[8] compared neoadjuvant pembrolizumab plus adjuvant pembrolizumab with adjuvant pembrolizumab alone in patients with resectable stage IIIB-D/IV cutaneous, acral, and mucosal melanomas. All patients received adjuvant pembrolizumab after surgery, and all patients received the same total number of cycles (18) of pembrolizumab; in the neoadjuvant arm, three cycles were given before surgery. Neoadjuvant pembrolizumab was associated with improved EFS. The 2-year EFS rate was 72% on the neoadjuvant followed by adjuvant arm and 49% on the adjuvant arm (difference, 23%; 95% CI, 11 to 35). OS was not reported, nor were EFS HRs.

#### Clinical Interpretation.

*Summary of the 2020 guideline.* At the time of the 2020 guideline, neoadjuvant therapy was considered investigational by the Expert Panel, and the Panel encouraged referral into neoadjuvant clinical trials. The Expert Panel recognized that core needle biopsy is the preferred diagnostic approach to clinically detected lymphadenopathy in any patient with known or suspected metastatic melanoma in regional nodes since excisional biopsy is more morbid and is not routinely required for either pathologic diagnosis or molecular characterization of melanoma.

*Updates for the 2023 guideline.* The SWOG 1801 trial[8] provides evidence in favor of neoadjuvant pembrolizumab. The reported benefit for EFS was large (23% fewer events at 2 years). Second, as all patients received the same total number of cycles of pembrolizumab, a regimen already recommended in this guideline for adjuvant therapy, there is little reason to believe that patients receiving the recommended therapy will experience any additional harm from delivering three cycles as neoadjuvant therapy. The 1,801 trial included only 23 patients with stage IV disease. The Expert Panel believed that patients with stage IV disease would benefit from being offered neoadjuvant therapy, but recognizes that the evidence for that subpopulation is sparse.

The Dummer 2021 trial[27] results suggest that neoadjuvant T-VEC may provide benefits to both EFS and OS for patients. However, this trial has aspects that make its results difficult to interpret. First, it is relatively small (150 total patients). Second, it was deemed to be at high risk of bias as the differences in surgery rates and adjuvant therapy between the arms could have led to differences in outcome and might be due to the open-label nature of the trial. Given these concerns, the Panel did not believe that T-VEC could be recommended as a routine option for neoadjuvant therapy.

There is one known ongoing phase III trial, NADINA (ClinicalTrials.gov identifier: NCT04949113), studying neoadjuvant nivolumab plus ipilimumab versus adjuvant nivolumab, but no known trial that is investigating neoadjuvant nivolumab as single-agent therapy with a similar design to SWOG S1801. The Expert Panel was unable to come to a clear consensus regarding whether nivolumab and pembrolizumab could be considered sufficiently equivalent that evidence for one agent could be treated as evidence of the other in the neoadjuvant setting. Therefore, no recommendation for or against nivolumab as neoadjuvant therapy was made. See the Discussion section for a broader treatment of this issue.

### Clinical Question 2

What adjuvant systemic therapy options, alone or in combination, have demonstrated clinical benefits in adults with resected (stage II, stage III, stage IV) cutaneous melanoma? Are there subpopulations of patients (ie, clinical features, biomarker status, lymph node dissection *v* sentinel lymph nodes) who benefit more or less from those options?

*Summary of Results for Question 2*

**Summary of the 2020 guideline.** Seventeen randomized trials[9-11,45-70] of adjuvant therapy in cutaneous melanoma are described in detail in the Data Supplement (Tables 2020-1 to 2020-5). Most of these trials were considered at low risk of bias, and those with some concerns were not considered sufficiently flawed to warrant a detailed comment. In addition, the individual patient data meta-analysis by Suciu et al[71] addressing the use of adjuvant interferon (IFN) versus observation or no IFN was included. See the 2020 guideline publication and Data Supplement (Tables 2020-1 to 2020-5) for additional details.

**Updates for the 2023 guideline.** Five new randomized trials of adjuvant therapy in resectable cutaneous melanoma were identified and are fully described in the Data Supplement (Tables 1-5): CheckMate 915,[30] IMMUNED,[12] KEYNOTE-716,[28] SWOG S1404,[29] and US Intergroup E1609.[31] None of these trials were considered at high risk of bias. In addition, the Panel was aware of a conference abstract[13] reporting results from CheckMate 76K.

*Recommendation 2.1.1*

Adjuvant pembrolizumab or nivolumab should be offered to patients with resected stage IIB or IIC melanoma. See Table 1 for reasonable doses and schedules (Type: Evidence based; Evidence quality: Moderate; Strength of recommendation: Strong).

**Qualifying statement.** The recommendation in favor of nivolumab is based at the time of writing on the conference abstract publication of CheckMate 76K[13] and under the assumption that full publication of that data will not contradict the abstract. See the Clinical Interpretation section for further details.

*Recommendation 2.1.2*

Adjuvant therapy should not be offered to patients with resected stage IIA melanoma outside of enrollment in a clinical trial (Type: Informal consensus; Evidence quality: Not applicable; Strength of Recommendation: Strong).

**Literature review and analysis.**

*Summary of the 2020 guideline.* No positive trials were identified that informed treatment in stage II disease.

*Updates for the 2023 guideline.* The KEYNOTE-716 trial[28] compared pembrolizumab with placebo in patients with completely resected stage IIB/C melanoma with no regional lymph node involvement. The median RFS was not yet reached in either arm, but the RFS HR was 0.61 (95% CI, 0.45 to 0.82) in favor of pembrolizumab. The twelve-month RFS was 90% with pembrolizumab versus 83% with placebo; the 18-month RFS was 86% versus 77%. Grade 3 or higher TRAEs were more frequent with pembrolizumab (16%) than with placebo (4%). OS data are not yet available.

The CheckMate 915 trial[30] compared nivolumab plus ipilimumab versus nivolumab plus placebo in patients with completely resected stage IIIB-D or IV melanoma. No significant difference in either RFS (HR, 0.92; 95% CI, 0.77 to 1.09) or OS (HR, 1.09; 95% CI, 0.80 to 1.32) was reported. Grade 3 and 4 TRAEs were more frequent with combination (32.6%) versus nivolumab alone (12.8%).

The CheckMate 76K trial[13] compared nivolumab with placebo in patients with completely resected stage IIB-C melanoma with no regional lymph node involvement. The median EFS was not reached in either arm, but the EFS HR was 0.42 (95% CI, 0.30 to 0.59) in favor of nivolumab. The twelve-month EFS was 89% with nivolumab versus 79% with placebo. Grade 3 or higher TRAEs were more frequent with nivolumab (10%) than placebo (2%). OS data are not yet available.

**Clinical interpretation.** For the 2020 guideline, the absence of data did not allow for a recommendation on adjuvant therapy for stage II melanoma. Since that time, the publication of the KEYNOTE-716 trial supports a recommendation in favor of pembrolizumab for patients with stage IIB-C melanoma. Although OS data are not yet reported, the RFS benefit found in the trial is substantial and reasonably certain. Adverse events from pembrolizumab are well understood and manageable.

The CheckMate 76K trial[13] supports a recommendation in favor of nivolumab in the same patient population although this is less clear as the data are only available in abstract form. This trial was very similar in design to KEYNOTE-716, except that nivolumab was used instead of pembrolizumab and assignment was 2:1 to nivolumab versus placebo. The Expert Panel decided to include nivolumab in the recommendation for several reasons. First, the reported data on EFS, RFS, and toxicity are very similar between the two trials. Second, because of the similarity in design, the Panel had little reason to believe that CheckMate 76K was of substantially lower evidence quality than KEYNOTE-716 although a full evaluation of the trial is not yet possible. Finally, the Panel weighed the potential costs of making a somewhat premature recommendation against the costs of not making such a recommendation and therefore likely ensuring that the recommendation would be out of date within a very short period of time. On the basis of this reasoning, a recommendation in favor of nivolumab was warranted.

While the consensus of the Panel was that a strong recommendation could be made for adjuvant nivolumab or pembrolizumab for these patients, the potential for serious morbidity, both acute and long term, because of immune-related toxicity must be considered and weighed against the lack of OS data in stage II disease. The ASCO guideline, *Management of Immune-Related Adverse Events in Patients Treated With Immune Checkpoint Inhibitor Therapy*,[72] has a full discussion of both the potential incidence of these toxicities and recommendations for managing and mitigating them.

Therefore, the previous recommendation against therapy has been amended as recommendation 2.1.2, and recommendation 2.1.1 was created for patients with stage IIB-C resected melanoma.

*Recommendation 2.2*

For patients with resected stage IIIA-D disease that is *BRAF* wild-type, the following adjuvant therapy options should be offered (in no particular order): nivolumab × 52 weeks OR pembrolizumab × 52 weeks. Ipilimumab and high-dose IFN are not recommended for routine use in adjuvant therapy. See the Clinical Interpretation section on shared decision making with patients between these options. See Table 1 for recommended dosing and scheduling details (Type: Evidence based, benefits outweigh the harms; Evidence quality: High; Strength of recommendation: Strong).

**Qualifying statements.** Patients with stage IIIA disease with microscopic sentinel nodal metastasis < 1 mm in diameter were not included in the randomized trials that studied efficacy of immune checkpoint inhibitors as adjuvant therapy for melanoma. Both nivolumab and pembrolizumab are US Food and Drug Administration (FDA)–approved adjuvant treatments for patients with melanoma with lymph node involvement who have undergone complete disease resection. Patients with stage IIIA disease with <1 mm involvement in the sentinel lymph node have a relatively better prognosis and lower risk of relapse. Therefore, treatment should be individualized after discussing risk-benefit quotient with these patients.

**Literature review and analysis.**

*Summary of the 2020 guideline.* One meta-analysis[71] and 14 randomized trials[9,10,49,50,52,54-56,59,60,63,64,66-68] in stage III melanoma were identified where *BRAF* mutation status was not an eligibility requirement. Of these, only three reported on therapies that were relevant to the recommendations: European Organisation for Research and Treatment of Cancer (EORTC) 18071,[61-63] EORTC 1325/KEYNOTE-054,[9] and CheckMate 238.[10] The remainder investigated IFN therapy or other therapies that are no longer relevant to recommendations and are not further discussed in this update. EORTC 18071 found significant improvements in RFS and OS for the cytotoxic T-lymphocyte-associated antigen 4 (CTLA-4)–blocking antibody ipilimumab versus placebo (RFS HR, 0.76; 95% CI, 0.64 to 0.89; $P < .001$, OS HR, 0.72; 95% CI, 0.58 to 0.88; $P = .001$). EORTC 1325/KEYNOTE-054 found significant improvements in RFS for the PD-1–blocking antibody pembrolizumab versus placebo (RFS HR, 0.57; 98.4% CI, 0.43 to 0.74; $P < .001$). CheckMate 238,[10] found that nivolumab had significantly improved RFS compared with ipilimumab (RFS HR, 0.65; 97.56% CI, 0.51 to 0.83; $P < .001$). See the 2020 guideline publication and Data Supplement (Tables 2020-1 to 2020-5) for additional details.

*Updates for the 2023 guideline.* Full results from the US Intergroup E1609 trial[31] are now available. These results are not meaningfully different from the conference abstract mentioned in the 2020 Guideline. The SWOG S1404 trial reported by Grossman et al[29] compared pembrolizumab with high-dose IFN or, after a protocol amendment, ipilimumab. Pembrolizumab was associated with significantly improved RFS (HR, 0.76; 99.62% CI, 0.59 to 0.99) but not OS (HR, 0.82, 96.3% CI, 0.61 to 1.09). Grade 3 or worse TRAEs were more frequent with IFN and ipilimumab than with pembrolizumab.

Two network meta-analyses, Longo et al[73] and Toor et al,[74] that calculated estimates of efficacy of pembrolizumab versus nivolumab (among other comparisons) by way of the indirect comparison of CheckMate 238 (nivolumab *v* ipilimumab) with EORTC 18071 (ipilimumab *v* placebo) with EORTC 1325 (pembrolizumab *v* placebo) were identified. As these meta-analyses used somewhat different methods, the estimates were not identical, but neither estimated a statistically significant difference between pembrolizumab versus nivolumab. Estimated RFS HRs were: Toor et al[74], HR, 1.08 (95% CI, 0.78 to 1.49) and Longo et al, HR, 1.28 (95% CI, 0.84 to 1.94). Longo et al also estimate the odds ratio of grade 3 or 4 adverse events of pembrolizumab versus nivolumab to be 2.13 (95% CI, 1.36 to 3.34). Note that in the study by Longo et al, in the abstract, it is indicated that the odds ratio was estimated for adverse events. However, in the Data Supplement (Figure 5, the figure that reports the data), the term hazard ratio was used. This is likely an error as HRs would only be appropriate for time-to-event outcomes. This value is reported as an odds ratio here.

**Clinical interpretation.** In the 2020 guideline, based primarily on the Suciu meta-analysis evidence, the Expert Panel determined that IFN could not be recommended. On the basis of the superiority of nivolumab to ipilimumab in CheckMate 238, adjuvant ipilimumab was not recommended as the preferred form of adjuvant immunotherapy. Nivolumab was recommended on the basis of the CheckMate 238 trial results, and pembrolizumab on the basis of EORTC 1325/KEYNOTE-054 trial results and the comparability of those results with CheckMate 238 results. Since 2020, the SWOG S1404 results support the previous results of the EORTC 1325 trial that pembrolizumab is a reasonable choice of therapy. The network meta-analyses suggest that there is insufficient evidence to prioritize the use of either pembrolizumab or nivolumab in this population. Therefore, no changes were made to recommendation 2.2.

### *Recommendation 2.3*

For patients with resected stage IIIA-D disease that is *BRAF*-mutant (V600E/K*), the following adjuvant therapy options should be offered (in no particular order): nivolumab × 52 weeks OR pembrolizumab × 52 weeks OR dabrafenib plus trametinib × 52 weeks. See the Clinical Interpretation section on shared decision making with patients between these options. See Table 2 for reasonable dosing and scheduling details (Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

**Qualifying statements.** Patients with stage IIIA disease with microscopic sentinel nodal metastasis <1 mm diameter were not included in the randomized trials that studied the efficacy of immune checkpoint inhibitors and BRAF-targeted agents as adjuvant therapy for melanoma. Nivolumab, pembrolizumab, and dabrafenib plus trametinib are FDA-approved adjuvant treatments for patients with melanoma with lymph node involvement who have undergone complete resection of their disease. Patients with stage IIIA disease with <1 mm involvement in the sentinel lymph node usually have a good prognosis and low risk of relapse. Therefore, treatment should be individualized after discussing risk-benefit quotient with these patients.

**Literature review and analysis.**

*Summary of the 2020 guideline.* In addition to the trials mentioned for recommendation 2.2, all of which included patients with *BRAF*-mutant melanoma, the COMBI-AD[11] trial of dabrafenib plus trametinib versus placebo was conducted specifically in patients with *BRAF* V600 E/K mutations and found significant improvements in RFS and OS. Subgroup analyses in the pembrolizumab KEYNOTE-054 and nivolumab CheckMate 238 trials of patients with resected melanoma found similar benefits for those with wild-type or *BRAF* V600 E/K–mutant disease.

*Updates for the 2023 guideline.* There were no new trials in this population. A network meta-analysis reported by Sharma et al[75] estimated the efficacy of dabrafenib plus trametinib versus pembrolizumab versus nivolumab in this population, with some caveats. As there is no way to directly link nivolumab to dabrafenib plus trametinib in a network of trials solely within this population, the network among the trials was CheckMate 238, *BRAF*-mutated subgroup analysis (nivolumab *v* ipilimumab); EORTC 18071, analysis of all patients (ipilimumab *v* placebo); COMBI-AD (dabrafenib plus trametinib *v* placebo); KEYNOTE-054, *BRAF*-mutated subgroup analysis (pembrolizumab *v* placebo). As this analysis made use of subgroups from some trials, the number of patients involved was limited. The estimated RFS HRs were dabrafenib plus trametinib versus nivolumab 0.88 (95% CI, 0.59 to 1.31), dabrafenib plus trametinib versus pembrolizumab 0.91 (95% CI, 0.57 to 1.44), and nivolumab versus pembrolizumab 1.03 (95% CI, 0.67 to 1.77).

**Clinical interpretation.** As no new data have been published since 2020, recommendation 2.3 was unchanged. The agents described in recommendation 2.2 are valid treatment options for *BRAF* V600E/K–mutant melanoma, but the results of the COMBI-AD trial indicate that dabrafenib plus trametinib is also a valid option for those patients and has preliminary evidence of an OS benefit. In the absence of head-to-head comparisons of the efficacy of these agents in this population, either (anti–PD-1 or dabrafenib plus trametinib) of these treatment options can be considered. This was supported by the published network meta-analyses. Severe (grade 3) treatment-related toxicity was reportedly higher with dabrafenib and trametinib (41%) as compared with nivolumab

(14.4%) or pembrolizumab (14.7%). In addition, the treatment discontinuation rate was higher in COMBI-AD (26%) than in CheckMate 238 (8%) or KEYNOTE-054 (10%).

*Recommendation 2.4*

No recommendation can be made for or against adjuvant BRAF/MEK inhibitor therapy in patients with resected stage III/IV melanoma with *BRAF* mutations other than V600E/K (Type: No Recommendation; Evidence quality: NA; Strength of recommendation: NA).

**Literature review and analysis.** No trials were identified in either the 2020 guideline or the 2022 update that studied adjuvant dabrafenib plus trametinib in patients with other *BRAF* mutations.

**Clinical interpretation.** In the absence of data on other *BRAF* mutations, BRAF/MEK inhibitor therapy cannot be specifically recommended for these patients.

*Recommendation 2.5.1*

Patients with resected stage IV melanoma should be offered adjuvant therapy (Type: Informal consensus; Evidence quality: Moderate; Strength of recommendation: Strong).

*Recommendation 2.5.2*

Reasonable options for therapy are: nivolumab alone (Evidence quality: Moderate), nivolumab plus ipilimumab followed by nivolumab (Evidence quality: Low), pembrolizumab alone (Evidence quality: Low), and dabrafenib plus trametinib (in patients with *BRAF* V600E/K disease; Evidence quality: Low). See the Clinical Interpretation section on shared decision making with patients between these options (Type: Informal consensus; Evidence quality: varies by option; Strength of recommendation: Weak).

**Literature review and analysis.**

*Summary of the 2020 guideline.* Only the CheckMate 238 trial of nivolumab versus ipilimumab included completely resected stage IV patients. No other data were available.

*Updates for the 2023 guideline.* The IMMUNED trial[12,76] compared the combination of nivolumab plus ipilimumab versus nivolumab plus placebo versus placebo in patients with stage IV melanoma and reported no evidence of disease after surgery and/or radiotherapy. The median OS was not reached for the combination, 43.2 months for nivolumab, and 31.9 months for placebo. The OS HRs were combination versus placebo 0.41 (95% CI, 0.17 to 0.99) and nivolumab versus placebo 0.75 (95% CI, 0.36 to 1.56). The median RFS was not reached with the combination, 12.4 months with nivolumab, and 6.5 months with placebo. The RFS HRs were combination versus placebo 0.23 (95% CI, 0.12 to 0.45), nivolumab versus placebo 0.56 (95% CI, 0.33 to 0.94), and combination versus nivolumab alone 0.40 (95% CI, 0.20 to 0.79). The twelve-month RFS was 75% (combination), 52% (nivolumab), and 32% (placebo). Grade 3 or greater TRAEs were most frequent with the combination (71%) and more frequent with nivolumab (27%) than with placebo (6%).

The subgroup analysis of patients with stage IV disease from the CheckMate 915 trial[30] is relevant; 245 patients with completely resected stage IV disease were included in that trial. RFS was not significantly different between nivolumab plus ipilimumab and nivolumab plus placebo (HR, 0.88; 95% CI, 0.58 to 1.32).

**Clinical interpretation.** The Expert Panel was able to achieve consensus to make a strong recommendation in favor of adjuvant therapy for patients with stage IV disease. However, the Panel was not able to reach consensus to strongly recommend a particular adjuvant therapy option over any other. In all cases, the evidence is either indirect, from subgroup analyses, or complicated by other matters.

- Nivolumab alone: The evidence from the CheckMate 238 trial melanoma was sufficient to recommend single-agent nivolumab. This would be considered moderate quality of evidence as only a subset of patients in that trial were stage IV.
- Nivolumab plus ipilimumab: The results from the IMMUNED trial[76] provide some evidence for this combination. This small phase II trial was deemed to be at low risk of bias and reported that nivolumab plus ipilimumab is associated with longer RFS than both placebo and single-agent nivolumab and, on its own, would be considered moderate quality evidence. Given all the available evidence (from IMMUNED, CheckMate 915, and the stage III trials), it is likely that nivolumab plus ipilimumab has the highest rate of serious adverse events of all the potential options. In addition, while the dose and schedule were not the same, the subgroup analysis in stage IV disease from CheckMate 915 did not find any improvement in RFS for nivolumab plus ipilimumab versus nivolumab and that subgroup (245 patients) was larger than the relevant arms of IMMUNED (115 patients). These mixed results, taken as a whole, were low-quality evidence in favor of the combination. The higher rate of adverse events should be taken into consideration by clinicians and patients when choosing the appropriate therapy.
- Pembrolizumab alone: Although pembrolizumab has not been investigated in these patients, given the similarities in effect between nivolumab and pembrolizumab found in the key trials, it was reasonable to recommend pembrolizumab as an alternative to nivolumab in resected stage IV patients. Given the indirectness of this evidence, it must be considered low quality.
- Dabrafenib plus trametinib: Given the data from the relevant trials of dabrafenib plus trametinib in stage III disease, the Expert Panel felt that it was reasonable to

recommend adjuvant dabrafenib plus trametinib in these patients when their disease is *BRAF* V600E/K–mutant. Given the indirectness of this evidence, it must be considered low quality. The Expert Panel felt that adjuvant dabrafenib plus trametinib may be of more value for patients who cannot be treated with, or cannot tolerate, adjuvant immunotherapy options.

*Other considerations with respect to question 2 from the 2020 guideline.* The 2020 guideline publication provided detailed discussion of various clinical presentations of stage III disease including microsatellite metastases, resection of locoregional recurrence after completion lymph node dissection, the shift to sentinel lymph node biopsy that has occurred since many of the important trials were conducted, the lower risk of recurrence in patients with stage IIIA disease, and appropriate molecular testing. That discussion is not replicated here.

### Clinical Question 3

What systemic therapy options, alone or in combination, have demonstrated clinical benefits in adults with unresectable and/or metastatic cutaneous melanoma? Are there subpopulations of patients (ie, clinical features, biomarker status, presence of brain metastases) who benefit more or less from those options?

*Summary of Results*

**Summary of the 2020 guideline**. The primary evidence base for this question used in the 2020 guideline is the Cochrane review.[22] The Cochrane review comprehensively identified RCTs of systemic therapy conducted in patients with metastatic melanoma up to October 2017 and included 122 RCTs that studied a wide range of systemic therapies and included multiple meta-analyses. Nine additional randomized phase III trials[21,77-85] that were not contained in the Cochrane review were identified, four of which were considered to have irrelevant comparisons and/or results[78,80,82,85] and five that were relevant to recommendations: Ascierto et al[77]; CheckMate 511,[79] COLUMBUS,[21,81] NEMO,[83] and OPTiM.[84] These trials are described in detail in the Data Supplement (Tables 2020-1 to 2020-5). All but the trial by Ugarel et al[85] were considered low risk of bias. These data are summarized as needed with each recommendation statement.

**Updates for the 2023 guideline**. Twelve new randomized trials[15,32-34,36-42,86] of therapy in unresectable or metastatic cutaneous melanoma were identified and are fully described in the Data Supplement (Tables 1-5). Of these, four trials are not considered relevant to any recommendations and are not further discussed: KEYNOTE-029,[36] Lebbé et al,[37] MIRACULUM,[38] and SWOG S1320.[42] The remainder are discussed in the relevant Literature Review and Analysis section.

Additional publications for trials previously identified, in either the 2020 guideline or the Cochrane review, are itemized in the Data Supplement (Tables 7 and 8). None were considered relevant to the recommendations.

*Recommendation 3.1*

For patients with *BRAF* wild-type, unresectable and/or metastatic cutaneous melanoma, the following treatment options should be offered (in no particular order): nivolumab plus ipilimumab followed by nivolumab OR nivolumab plus relatlimab OR nivolumab OR pembrolizumab. See Table 2 for recommended dosing and scheduling details. See the Clinical Interpretation section on shared decision making with patients between these options (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

**Literature review and analysis.**

*Summary of the 2020 guideline.* The meta-analyses of the Cochrane review provided key evidence for trials published before 2017. Anti–PD-1 therapy was associated with statistically significant improvements in OS (HR, 0.63; 95% CI, 0.60 to 0.66; one trial) and progression-free survival (PFS; HR, 0.54; 95% CI, 0.50 to 0.60; two trials) compared with anti–CTLA-4 therapy, as well as statistically significantly reduced grade 3 and 4 toxicities (relative risk [RR], 0.70; 95% CI, 0.54 to 0.91; two trials). These results were driven by the CheckMate 067[14,87] trial of nivolumab plus ipilimumab versus nivolumab alone versus ipilimumab alone, and the KEYNOTE-006[16] trial of pembrolizumab versus ipilimumab. The combination of anti–PD-1 therapy plus anti–CTLA-4 therapy was associated with statistically significant improvements in PFS (HR, 0.40; 95% CI, 0.35 to 0.46; two trials) and significantly increased grade 3 and 4 toxicities (RR, 1.57; 95% CI, 0.85 to 2.92; two trials) compared with anti–CTLA-4 therapy alone. In addition, two other randomized trials that were potentially relevant to recommendations were identified in the 2020 guideline: CheckMate 511[79] studied an alternate dosing schedule of nivolumab and ipilimumab, and Ascierto et al[77] studied higher-dose single-agent ipilimumab.

*Updates for the 2023 guideline.* Two new randomized trials[15,33] in patients with *BRAF* wild-type melanoma and considered relevant to recommendations were published with full details in the Data Supplement (Tables 1-5).

The RELATIVITY-047 trial[15] compared the combination of relatlimab and nivolumab with nivolumab alone in previously untreated patients with unresectable or metastatic stage III-IV melanoma. The median PFS was 10.12 months with the combination versus 4.63 months with single-agent nivolumab. The PFS HR was 0.75 (95% CI, 0.62 to 0.92). The estimated 12-month PFS rate was 47.7% versus 36.0%. OS is not yet reported. Grade 3 or 4 TRAEs were more frequent with the combination (18.9%) than single-agent (9.7%) therapy. The IMSpire170 trial[33] compared the combination of cobimetinib and atezolizumab with single-agent pembrolizumab in patients with *BRAF* wild-type unresectable or metastatic melanoma. No significant differences in PFS or OS were found.

An exploratory network meta-analysis was conducted, fully described in the Data Supplement 2, to investigate whether the available evidence was sufficient to prioritize between the different options for patients with *BRAF*-mutant unresectable and/or metastatic melanoma. This analysis suggests the following:

- In general, two drug combinations (nivolumab plus ipilimumab, relatlimab plus nivolumab) have better estimated PFS than single drugs alone (pembrolizumab, nivolumab)
- Nivolumab plus ipilimumab has the highest estimated serious adverse event frequency, and nivolumab alone has the lowest.

However, given the limitations of this analysis described in the Data Supplement 2, none of these points can be considered as high certainty evidence on which to base recommendations and are therefore areas for future research.

**Clinical interpretation.** In the 2020 guideline, single-agent nivolumab and single-agent pembrolizumab were recommended as reasonable treatment options on the basis of data from two trials, CheckMate 067[14] and KEYNOTE-006[16] trials. Both trials reported significant OS benefit for anti–PD-1 therapy over ipilimumab. In addition, although the 067 trial was not powered for a comparison between the combination versus nivolumab monotherapy, the trial found increased PFS for the combination of ipilimumab and nivolumab compared with nivolumab alone, but the OS difference was not statistically significant, and the combination also had significantly increased toxicity. Therefore, the combination of ipilimumab and nivolumab was considered a reasonable treatment option in patients willing to accept the likely increased toxicity of the combination. Similarly, the reported PFS benefit for nivolumab plus relatlimab in RELATIVITY-047,[15] published in 2022, provides sufficient evidence to recommend combination as an additional reasonable option for therapy. In the 2020 guideline, the alternate dosages studied in CheckMate 511[79] (for the combination of nivolumab and ipilimumab) and Ascierto 2017[77] (for single-agent ipilimumab) were not recommended. Imspire170[33] found no benefits for atezolizumab plus cobimetinib compared with pembrolizumab.

Therefore, there are four reasonable therapy options for patients with *BRAF* wild-type unresectable or metastatic melanoma: nivolumab plus ipilimumab, nivolumab plus relatlimab, single-agent nivolumab, and single-agent pembrolizumab. In deciding between these options, several factors can be considered. First, combination therapy is associated with a greater PFS benefit than single-agent nivolumab, but with increased toxicity. Although there is no direct trial comparison, the exploratory meta-analysis and the similarity in results across many trials in benefits and harms between these anti–PD-1 agents suggest that this is likely the case with single-agent pembrolizumab as well. For patients with brain metastases, the joint ASCO, Society for Neuro-Oncology (SNO), and American Society for Radiation Oncology (ASTRO) brain metastases guideline[88] recommended nivolumab plus ipilimumab as the most reasonable option for systemic therapy, on the basis of results from a phase II trial[89] of nivolumab plus ipilimumab in asymptomatic brain metastases. Finally, in all cases, the decision on the choice of therapy should be made jointly with the patient after considering the risk and benefits; patients who prefer more aggressive therapy may wish to receive combination therapy.

A qualifying statement in the 2020 guideline noted that in the relevant randomized trials, nivolumab could be continued beyond 2 years,[14] whereas pembrolizumab was limited to 2 years. It is possible that shorter courses of therapy, as little as 1 year, may be reasonable. However, no high-quality data in the melanoma setting address what the duration of therapy should be. The qualifying statement also noted that for longer dosing cycles (eg, up to 6 weeks), appropriate monitoring for toxicity and disease progression is still necessary.

*Recommendation 3.2.1*

For patients with *BRAF* mutant (V600) unresectable and/or metastatic cutaneous melanoma, one of the following treatment options should be offered as first-line therapy: nivolumab plus ipilimumab followed by nivolumab OR nivolumab plus relatlimab OR nivolumab OR pembrolizumab OR dabrafenib plus trametinib OR encorafenib plus binimetinib OR vemurafenib plus cobimetinib. See Table 2 for recommended dosing and scheduling details. See the Clinical Interpretation section on shared decision making with patients between these options (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

*Recommendation 3.2.2*

Combination therapy with nivolumab plus ipilimumab is preferred as first-line therapy over BRAF/MEK inhibitor combination therapy. See the Clinical Interpretation section on shared decision making with patients between these options (Type: Evidence based; Evidence quality: High for dabrafenib plus trametinib, Low for other BRAF/MEK inhibitors; Strength of recommendation: Strong).

**Literature review and evidence.**

*Summary of the 2020 guideline.* In addition to the data described for recommendation 3.1 for patients where *BRAF* mutation status was not considered, the Cochrane review meta-analysis identified a subset of trials in patients with *BRAF*-mutant metastatic melanoma. The combination of BRAF and MEK inhibitor therapies was associated with statistically significant improvements in OS (HR, 0.70; 95%

CI, 0.59 to 0.82; four trials) and PFS (HR, 0.56; 95% CI, 0.44 to 0.71; four trials) compared with BRAF inhibitor therapy alone. Grade 3 and 4 toxicities were similar (RR, 1.01; 95% CI, 0.85 to 1.20; four trials). These results were driven by the coBRIM trial[19] of vemurafenib plus cobimetinib versus vemurafenib alone and the COMBI-D trial,[90] COMBI-V trial,[20,91] and the trial by Flaherty et al[92] investigating dabrafenib plus trametinib. In addition to the trials identified in the Cochrane review, the COLUMBUS trial[21,81] compared encorafenib plus binimetinib versus encorafenib alone or vemurafenib alone. This trial found significant improvements in PFS of the combination versus vemurafenib (HR, 0.54; 95% CI, 0.41 to 0.71) but not versus encorafenib (HR, 0.75; 95% CI, 0.56 to 1.00). A significant difference in OS between the combination and vemurafenib (HR, 0.61; 95% CI, 0.47 to 0.79) was found. Grade 3 and 4 toxicities were reported at similar rates among the three arms.

*Updates for the 2023 guideline.* In addition to the RELATIVITY-047 trial[15] described under recommendation 3.1, which included patients regardless of *BRAF* status, four new randomized trials[32,34,42,86] were reported in patients with *BRAF*-mutated melanoma. Of these, the SWOG S1320 trial[42] compared an intermittent schedule of dabrafenib plus trametinib with the standard continuous schedule. It found no benefits for the intermittent schedule and is not discussed further.

The remaining three trials all investigated the addition of a PD-L1 inhibitor to the combination of BRAF/MEK inhibitor therapy. The Imspire150 trial[32] compared atezolizumab with placebo in patients who were also receiving vemurafenib and cobimetinib. This trial reported a significant difference in PFS, HR 0.78 (95% CI, 0.63 to 0.97), but not for OS, HR 0.85 (95% CI, 0.64 to 1.11). Note that per protocol, the primary outcome was PFS as assessed by the investigator. PFS was not different between the arms when assessed by independent review (HR, 0.85; 95% CI, 0.67 to 1.07). Similar rates of severe toxicity were reported on each arm. However, 22 patients originally assigned to the triplet combination arm were discontinued on therapy during cycle 1 before receiving atezolizumab and were analyzed as part of the safety population on the placebo arm instead of the atezolizumab arm. This potentially biased toxicity rates in favor of the atezolizumab arm. The KEYNOTE-022 trial compared pembrolizumab with placebo in patients who were receiving dabrafenib plus trametinib and reported no significant difference in PFS, HR 0.66 (95% CI, 0.40 to 1.07). A follow-up publication of KEYNOTE-022[35] reported PFS HR 0.58 (95% CI, 0.34 to 0.83) and OS HR 0.64 (95% CI, 0.38 to 1.06) for pembrolizumab compared to placebo. However, given the design of the trial, the authors did not consider the PFS HR to be a statistically significant result. The COMBI-I trial compared spartalizumab versus placebo in patients who were receiving dabrafenib plus trametinib and reported no significant difference in PFS, HR 0.82 (95% CI, 0.66 to 1.03), or OS, HR 0.79 (95% CI, 0.59 to 1.05).

Two other relevant trials were published since 2020 with similar designs that studied BRAF plus MEK inhibition first with nivolumab plus ipilimumab after progression or vice versa. The DREAMseq trial[40] studied dabrafenib plus trametinib. As proportional hazards were not expected, this trial is unique among identified trials in that its primary outcome was 2-year OS rate and not PFS or OS HR. The two-year OS rate was 71.8% in patients who received nivolumab plus ipilimumab first versus 51.5% for those who received dabrafenib plus trametinib first. The difference in OS was significant (log-rank $P = .01$). The median PFS was 11.8 months for nivolumab plus ipilimumab as first line versus 8.5 months for dabrafenib plus trametinib (log rank $P = .054$). The SECOMBIT trial[41] studied encorafenib plus binimetinib. This trial was designed to be noncomparative; HRs and their confidence intervals were considered exploratory. It included a sandwich arm that provided 8 weeks of encorafenib plus binimetinib before nivolumab plus ipilimumab as first-line therapy, which is not further discussed. No significant differences in PFS (HR, 0.71; 95% CI, 0.44 to 1.14) or OS (HR, 0.73; 95% CI, 0.42 to 1.26) were reported.

The exploratory network meta-analysis described in the Data Supplement 2 also investigated whether the available evidence was sufficient to prioritize between the different options for patients with *BRAF*-mutant unresectable and/or metastatic melanoma. This analysis suggested that

- Addition of PD-L1 inhibitor to the combination of BRAF/MEK inhibitors might have better PFS but worse serious adverse event frequency than BRAF/MEK inhibitors alone, single-agent nivolumab or pembrolizumab, and two-drug nivolumab plus ipilimumab or nivolumab plus relatlimab.
- BRAF/MEK inhibitor combinations might have higher serious adverse event frequency than single-agent nivolumab, single-agent pembrolizumab, or nivolumab plus relatlimab, but not nivolumab plus ipilimumab.
- BRAF/MEK inhibitor combinations cannot be differentiated on either PFS or adverse events.
- Single-agent nivolumab not only may provide the lowest PFS of all the analyzed options but also might have the lowest frequency of serious adverse events.

However, given the limitations of this analysis described in the Data Supplement 2, none of these points can be considered as good evidence on which to base recommendations and are therefore areas for future research.

**Clinical interpretation.** Three combinations of BRAF and MEK inhibitors had been shown to be superior to BRAF inhibitor alone in the 2020 guideline: vemurafenib plus cobimetinib in CoBRIM,[19] dabrafenib plus trametinib in COMBI-D[90] and COMBI-V,[20,91] and encorafenib plus binimetinib in COLUMBUS[21,81] trial investigating encorafenib plus binimetinib. Therefore, in addition to the options described in recommendation 3.1, which now include nivolumab plus relatlimab, these three studied regimens were considered reasonable options for patients with *BRAF* V600E/K–mutant melanoma.

The COMBI-I, KEYNOTE-022, and Imspire150 trials[32,34,86] provide evidence regarding the addition of a PD-(L)1 inhibitor to BRAF/MEK inhibitor combination therapy, but the interpretation of this evidence is unclear. Two trials found potential benefits to PFS, but in both cases, those benefits are muddied by other factors. In Imspire150, the discrepancy between investigator- and independently assessed PFS suggests a potential for bias. In KEYNOTE-022, the benefit was only seen after extended follow-up and the authors state that the trial was not intended to be comparative. The Panel recognizes that the US FDA approved atezolizumab for this indication in 2020 on the basis of Imspire150. However, the Expert Panel decided that on the basis of the limited evidence for benefits and the likely increased toxicity that no combination of PD-(L)1 inhibitor with BRAF/MEK inhibitors should be recommended.

The complicated picture for *BRAF* wild-type melanoma becomes even more complicated for patients with *BRAF*-mutant melanoma; there are seven strongly recommended options that form the new recommendation 3.2.1. The evidence available to prioritize among these options is mixed and difficult to interpret. DREAMseq[40] reported that a strategy of nivolumab plus ipilimumab first-line followed by dabrafenib plus trametinib after progression had better 2-year OS than the reverse strategy. While RELATIVITY-047 reported better PFS for nivolumab plus relatlimab compared with nivolumab alone, it is unclear how this would compare with BRAF/MEK inhibitor combination therapy. The exploratory network meta-analysis did not provide sufficient information to prioritize among the available regimens. The Expert Panel found it difficult to achieve consensus on this matter as some members believed that a list of options was all that could be provided, whereas others believed that evidence from DREAMseq supported a strong recommendation in favor of nivolumab plus ipilimumab as the best choice for first-line therapy. Ultimately, the consensus was that a strong recommendation to prefer nivolumab plus ipilimumab over BRAF/MEK inhibitor therapy could be made, on the basis of the high-quality evidence of DREAMseq in the case of dabrafenib plus trametinib and extrapolating that result to the other BRAF/MEK inhibitor combinations as low-quality evidence given that SECOMBIT trial of encorafenib plus binimetinib was designed to be noncomparative and there is not a similar trial of vemurafenib plus cobimetinib. As noted in recommendation 3.1's Clinical Interpretation, the ASCO-SNO-ASTRO joint brain metastases guidelines[88] recommended nivolumab plus ipilimumab in patients with brain metastases. In addition, patients who are better able to tolerate toxicity and/or have better performance status may prefer combination therapy over single-agent anti–PD-1 therapy although this does not necessarily help in the decision between nivolumab plus relatlimab compared with BRAF/MEK inhibitor combination. As always, the decision on the choice of therapy should be made jointly with the patient after considering the risks and benefits.

In the 2020 guideline, a qualifying statement suggested that switching between BRAF/MEK inhibitor combinations may be reasonable if patients experience toxicity as each combination can present somewhat different toxicity profiles. In the clinical context of BRAF/MEK inhibitor failure, no prospective data exist regarding the efficacy of switching to a different BRAF/MEK combination. The statement also suggested that for longer dosing cycles for anti–PD-1 regimens (eg, up to 6 weeks), appropriate monitoring for toxicity and disease progression is still necessary.

*Recommendation 3.3*

After progression on anti–PD-1–based therapy, patients with unresectable and/or metastatic *BRAF* wild-type cutaneous melanoma may be offered ipilimumab or ipilimumab containing regimens. Patients should be offered or referred for enrollment in clinical trials where possible (Type: Informal consensus; Evidence quality: Low; Strength of recommendation: Weak).

**Literature review and analysis.** No trials were identified in either the 2020 guideline or the current update that addressed second-line or greater therapy in patients with *BRAF* wild-type unresectable and/or metastatic melanoma previously treated with the regimens recommended in recommendation 3.1 in first-line therapy. The Cochrane review identified one trial, reported by Hodi et al,[93] which investigated ipilimumab in second-line therapy. However, in that trial, patients were previously treated with chemotherapy or interleukin-2 (IL-2) and not anti–PD-1 therapy, so it is unclear how it is relevant to practice on the basis of the recommendations in this guideline and is not further discussed.

**Clinical interpretation.** Ipilimumab has been associated with improved OS and manageable toxicity in patients who have been treated with chemotherapy or IL-2 compared with a vaccine control. However, assuming that recommendation 3.1 are followed, patients will receive first-line anti–PD-1 therapy, potentially in combination with either ipilimumab or relatlimab. Therefore, the existing evidence is very limited as to the efficacy of ipilimumab after relapse on the recommended initial therapy regimens. However, the consensus of the Expert Panel was that, in the absence of direct trial evidence, ipilimumab- or ipilimumab-containing regimens could be offered to patients. It should be noted that ipilimumab at 3 mg/kg once every 3 weeks for up to four doses has been approved by the US FDA in these patients on the basis of the trial by Hodi et al.[93] In addition, ipilimumab-containing regimens could potentially include nivolumab plus ipilimumab in patients who have progressed on single-agent nivolumab or pembrolizumab. In response to external review feedback, a statement encouraging referral to clinical trials where possible was added to the recommendation.

The Panel notes that the results of the M14TIL trial[94] are relevant to this recommendation. In this trial, patients who had previous systemic treatment were randomly assigned to receive tumor-infiltrating lymphocytes (TILs) versus ipilimumab. TILs were not part of the scope of work for this guideline, so this trial was not included in the systematic review. Because of the nature of TIL therapy, patients had to have 2-3 cm total diameter of lesions that could be surgically resected to generate the TILs. All but 4 of the 168 included patients had previously received anti–PD-1 therapy, 40 as adjuvant therapy and 105 as first-line metastatic therapy. Patients receiving TILs had significantly improved PFS compared with those receiving ipilimumab (7.2 months *v* 3.1 months, HR, 0.50; 95% CI, 0.35 to 0.72). No significant difference in OS was reported (median 25.8 months with TILs *v* 18.9 months with ipilimumab, HR, 0.83; 95% CI, 0.54 to 1.27). The grade 3 or worse TRAE rate was 100% on TIL and 57% on ipilimumab. Patients on TILs reported generally better health-related quality-of-life scores.

In 2020, the Panel had recommended T-VEC as a possible option in the patients addressed by recommendation 3.3. At the time, this was based on the indirect data from the OPTIM trial.[84,95-97] However, given the additional indirect evidence from MASTERKEY-265,[39] the Panel no longer believed that T-VEC could be considered a reasonable option and this recommendation was dropped. Both these trials are addressed more fully under recommendation 3.5.

*Recommendation 3.4*

After progression on first line anti–PD-1–based therapy, patients with *BRAF* mutant (V600) unresectable and/or metastatic cutaneous melanoma may be offered combination BRAF/MEK inhibitor therapy as described in recommendation 3.2.1. Similarly, those who have progressed after combination BRAF/MEK inhibitor therapy may be offered anti–PD-1–based therapy as described in recommendation 3.2.1 (Type: Informal consensus; Evidence quality: Low; Strength of recommendation: Weak).

**Literature review and analysis.** In the 2020 guideline, only two randomized trials were identified of second-line or greater therapy that included patients with *BRAF*-mutant unresectable and/or metastatic melanoma who had received one of the options for first-line therapy in recommendation 3.2.1. The KEYNOTE-002 trial[17] of pembrolizumab versus chemotherapy did not exclude this population, but only included 125 patients with *BRAF*-mutant disease. It found a PFS benefit in the prespecified subgroup analysis for 10 mg/kg once every three weeks pembrolizumab (HR, 0.44; 95% CI, 0.26 to 0.74) but not 3 mg/kg once every three weeks pembrolizumab (HR, 0.74; 95% CI, 0.46 to 1.18). It was not clear what proportion of these patients had received BRAF/MEK inhibitor therapy versus ipilimumab-based therapy. The KEYNOTE-006[98] trial also included patients who had *BRAF*-mutant disease and previous therapy with BRAF/MEK inhibitor combination. In the subgroup analysis, a PFS benefit for pembrolizumab every 2 weeks versus ipilimumab was found in patients who had received previous BRAF inhibitor therapy (HR, 0.58; 95% CI, 0.34 to 0.97). No benefits were found with pembrolizumab every 3 weeks. The CheckMate 037 trial[99] did include second-line or greater patients with *BRAF*-mutant melanoma, but it required those patients to have progressed on anti–CTLA-4 and anti-BRAF therapy, so it is not clear if any patients received an option that is recommended in 3.2. Since the 2020 guideline, the DREAMseq and SECOMBIT trials have been published and are discussed in the literature review section for recommendation 3.2.

**Clinical interpretation.** In the absence of direct trial evidence other than small subgroup analyses, the consensus of the Expert Panel in 2020 was that it was reasonable to offer patients with *BRAF*-mutant disease systemic therapy on relapse on the basis of the indirect evidence of its efficacy in first-line therapy and no known evidence of cross-resistance between these therapies. The DREAMseq and SECOMBIT trials do not directly address this population because those trials addressed a strategy of first-line and second-line therapies and did not randomly assign a population that had all received the same first-line therapy to different second-line options. However, given the evidence in those trials of activity of BRAF/MEK after anti–PD-1 and in the reverse case, the Panel no longer believed that ipilimumab and ipilimumab-containing regimens were reasonable options after progression on first-line therapy, and the recommendation was altered to drop that option. While it is not directly mentioned in recommendation 3.4, the addition of nivolumab plus relatlimab to recommendation 3.2 makes that combination a potential option for patients who have progressed on first-line BRAF/MEK inhibitor combination therapy.

*Recommendation 3.5*

For patients with injectable (cutaneous or subcutaneous or nodal) unresectable lesions who are not eligible or do not desire the recommended systemic therapies, T-VEC may be offered as primary therapy (Type: Evidence based, benefits outweigh harms; Evidence quality: Low; Strength of recommendation: Weak).

**Literature review and analysis.** Two trials have been identified in the original 2020 guideline and for this update. The OPTIM trial[84,95-97] compared T-VEC with granulocyte-macrophage colony-stimulating factor in patients with unresectable and/or metastatic melanoma with injectable lesions. The median time to treatment failure (HR, 0.42; 95% CI, 0.32 to 0.54) was significantly improved, whereas OS (HR, 0.79; 95% CI, 0.62 to 1.00) was not. Grade 3 and 4 toxicities were somewhat increased. Chesney et al[80] compared T-VEC plus ipilimumab with ipilimumab alone and reported no significant difference in PFS (HR, 0.83; 95% CI, 0.56 to 1.23; *P* = .35) or OS (HR, 0.80; 95% CI, 0.44 to 1.46). These trials are fully described in the Data Supplement (Supplement 3, Tables 2020-1 to 2020-5). The MASTERKEY-265 trial[39] was identified in this full update, compared T-VEC with placebo in patients receiving single-agent pembrolizumab, and reported no significant difference in (HR, 0.86; 95% CI, 0.71 to 1.04) or OS (HR, 0.97; 95% CI, 0.77 to 1.21). It is fully described in the Data Supplement (Supplement 1, Tables 1-5). Given that the MASTERKEY-265 trial compared T-VEC with an option that has been recommended in this guideline, the evidence quality for recommendation 3.5 was downgraded from moderate to weak.

**Clinical interpretation.** Given the strength of the evidence identified for the options found in recommendations 3.1 and 3.2 and the infrastructure requirements necessary to deliver T-VEC therapy, the Expert Panel in the 2020 guideline felt that it was not possible to list T-VEC as an equivalent option solely on the basis of the OPTIM trial and in light of the trial by Chesney et al. However, in patients in whom those therapy regimens are contraindicated or the patients who are otherwise ineligible, or where the patient refuses the recommended regimens, T-VEC is a reasonable alternative assuming that the patient has injectable lesions. The lack of any PFS or OS benefit reported in MASTERKEY-265 reinforces this previous consensus.

**Other considerations.** Outside of these recommendations, clinicians should consider enrollment in clinical trials, where possible, before resorting to providing older therapies not recommended by this guideline (eg, IL-2) in second-line or greater settings. At this time, there are no randomized data to guide the choice of therapy in patients who have progressed on all the therapy options described in recommendations 3.1, 3.2.1, 3.3, and 3.4.

*NTRK fusions and tumor agnostic indications.* The US FDA has approved two agents, entrectinib (https://www.accessdata.fda.gov/drugsatfda_docs/label/2019/212725s000lbl.pdf) and larotrectinib (https://www.accessdata.fda.gov/drugsatfda_docs/label/2018/211710s000lbl.pdf), in patients with unresectable and/or metastatic solid tumors harboring *NTRK* fusions and who have progressed on standard therapies and/or who have no other viable alternative treatments. The approved indications would include patients with unresectable and/or metastatic melanoma whose tumors had progressed on therapy as recommended in 3.1, 3.2, 3.3, and 3.4. Therefore, it may be reasonable to offer testing for *NTRK* fusion in these patients and offer one of these therapies if the fusion is found. However, in the decision making around offering this therapy, the prevalence of NTRK fusion in melanoma should be considered, as stated in recommendation 3.2.1 of the recent ASCO provisional clinical opinion on somatic genetic testing.[100] That opinion reports an estimated 0.2% prevalence of *NTRK* fusion in patients with cutaneous melanoma. In addition, the nonrandomized studies on which these approvals were based included very few patients with melanoma. The Panel does note that a higher prevalence of *NTRK* fusions has been found in spitzoid melanoma; see the review by Forschner et al[101] for further information. Clinicians are encouraged to read the ASCO provisional clinical opinion[100] for further guidance on appropriate use of somatic genetic testing.

*Brain metastases.* ASCO, SNO, and ASTRO have jointly published a guideline[88] on the treatment of patients with brain metastases, which will include patients with melanoma. Recommendation 2.6. of that guideline states "Ipilimumab plus nivolumab (for all patients regardless of *BRAF* status) or dabrafenib plus trametinib (for patients with *BRAF* V600E mutation) may be offered to patients with asymptomatic brain metastases from melanoma. If these agents are used, local therapy may be delayed until there is evidence of intracranial progression (Type: Informal consensus; Evidence quality: Low; Strength of recommendation: Weak)." In addition, ASTRO has published a radiation therapy–specific guideline,[102] which has been endorsed by ASCO.[103] These guidelines provide recommendations that should be followed for patients with brain metastases from melanoma. While this subpopulation was not formally included in the updated systematic review for this guideline, no substantial new evidence published after the systematic review in the joint ASCO-SNO-ASTRO guideline was noted.

*Immune-related toxicities and autoimmune disease.* When immunotherapy is given, clinicians should maintain a high suspicion for immune-related toxicity. ASCO has published a comprehensive guideline[72] on the identification and management of this toxicity that should be used for patients with melanoma receiving these therapies.

*Performance status*. Performance status, although important, is not as relevant a consideration in the choice of therapy for systemic treatment of melanoma as it would be with other cancer treatment regimens (eg, chemotherapy). If poor performance status is primarily due to tumor burden, there is potential for improvement of performance status with the initiation of systemic therapy, whether with immunotherapy or targeted therapy. If poor performance status is due to comorbidities, careful consideration of the potential risks of therapy is warranted.

*Testing.* As noted in the Other Considerations section for question 2, *BRAF* mutation testing is an important component of the selection of therapy for patients with melanoma as the recommended therapy differs depending on *BRAF* mutation status. While the potential exists that other testing may be of high value in the future, at this time, the Expert Panel could not make any formal recommendation in favor of specific tests. With respect to patients with *BRAF* wild-type disease that progresses, it is technically possible that previous testing resulted in a false negative.[104-106] Therefore, retesting *BRAF* status on progression could be considered, which, however, may rarely be of value as the Expert Panel is not aware of any high-quality data that support this retesting. Finally, *NRAS* and *KIT* mutations have been published as driver mutations, particularly in melanomas occurring within sun-damaged skin.[107,108] The Expert Panel did not feel that data regarding the efficacy of inhibitors targeting these pathways were sufficient to make a formal recommendation regarding the testing of these two genes. Clinicians are encouraged to review ASCO's provisional clinical opinion on somatic genetic testing[100] for a broader discussion of when to test in a context where direct randomized evidence for therapy associated with a driver mutation is available.

### Clinical Question 4

What systemic therapy options, alone or in combination, have demonstrated clinical benefits in adults with noncutaneous melanoma (stage II or greater)? Are there subpopulations of patients (ie, clinical features, biomarker status, specific type of melanoma) who benefit more or less from those options?

*Recommendation 4.1.1*

HLA-A*02:01-positive patients with metastatic uveal melanoma should be offered tebentafusp (20 μg once on day 1, 30 μg once on day 8, and 68 μg once weekly after that until progression; Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

*Recommendation 4.1.2*

For all patients with uveal melanoma other than those addressed by recommendation 4.1.1, no recommendation for or against any specific systemic therapy may be made at this time. Patients should be offered or referred for enrollment in clinical trials where possible (Type: No recommendation; Evidence quality: Very Low; Strength of recommendation: Not applicable).

**Literature review and analysis.** Three trials of patients with uveal melanoma have been identified in the 2020 guideline and this update. Carvajal 2014[109,110] compared selumetinib with temozolomide or dacarbazine (investigator's choice) in patients with metastatic uveal melanoma and reported significantly improved PFS (HR, 0.46; 95% CI, 0.30 to 0.71) with selumetinib but no difference in OS (HR, 0.66; 95% CI, 0.41 to 1.06). More grade 3 toxicity was reported with selumetinib. However, in the follow-up phase III SUMIT trial[111] of selumetinib and dacarbazine versus placebo and dacarbazine, neither PFS (HR, 0.78; 95% CI, 0.48 to 1.27; $P = .32$) nor OS (HR, 0.75; 95% CI, 0.39 to 1.46; $P = .40$) were found to differ significantly between the arms. The EORTC 18021 trial[112] compared fotemustine delivered by hepatic intra-arterial (HIA) catheter versus intravenously in patients with uveal melanoma and liver metastases. It was stopped early because of an unplanned futility analysis at 171 patients. The trial reported significantly greater PFS with HIA (HR, 0.62; 95% CI, 0.45 to 0.84) but no difference in OS (HR, 1.09; 95% CI, 0.79 to 1.50). These trials are fully described in the Data Supplement (Supplement 3, Tables 2020-1 to 2020-5).

The study by Nathan 2021[43] was identified in this full update and is fully described in the Data Supplement (Supplement 1, Tables 1-5). It compared tebentafusp with the investigator's choice of pembrolizumab, ipilimumab, or dacarbazine in previously untreated, HLA-A*02:01–positive patients with metastatic uveal melanoma. Significant improvements in both OS, HR 0.51 (95% CI, 0.37 to 0.71), and PFS, HR 0.73 (95% CI, 0.58 to 0.94), were reported. This trial prompted a rapid recommendation update to the 2020 guideline[24]; this update supersedes that previous rapid recommendation article.

**Clinical interpretation.** At the time of the 2020 guideline, no recommendation could be made for or against any specific therapy for uveal melanoma. The publication of Nathan 2021, however, prompted a rapid recommendation article, and the new recommendations are presented here in this full update. Tebentafusp in HLA-A*02:01–positive patients with metastatic uveal melanoma is associated with a statistically significant increase in 1-year OS and 6-month PFS. While the frequency of grade 3 and 4 adverse events is substantial with tebentafusp, toxicity becomes more manageable with time and does not outweigh the observed benefits. The higher disease control rate with tebentafusp compared with the control arm may be responsible for the improved landmark PFS and improved OS in this study despite the low objective response rate. Further studies to understand the relationship between clinical benefit from tebentafusp or similar agents in advanced uveal melanoma and the spectrum of traditional tumor response criteria in solid tumors are required.

In the rapid update published in 2022, recommendation 4.1.1 was limited to previously untreated patients, on the basis of the inclusion criteria in the study by Nathan 2021. However, for this full update, the Expert Panel believed that was a needless restriction that might unnecessarily prevent patients from accessing the therapy and dropped that restriction.

*Recommendation 4.2*

In the absence of further data, the consensus of the Expert Panel is that patients with unresectable and/or metastatic mucosal melanoma may be offered therapy as described in recommendations 3.1 through 3.5. Patients should be offered or referred for enrollment in clinical trials where possible (Type: Informal consensus; Evidence quality: Very Low; Strength of recommendation: Weak).

**Literature review and analysis.** One trial, by Lian et al,[113] was identified in the 2020 guideline that was specifically conducted in mucosal melanoma. This phase II trial included patients with resected mucosal melanoma and studied adjuvant therapy with three arms: observation (surgery alone), high-dose IFN, and temozolomide plus cisplatin. It found statistically significant improvements in both relapse-free survival (chemotherapy 20.8 months *v* IFN 9.4 months,

$P$ < .001) and OS (chemotherapy 48.7 months *v* IFN 40.4 months, $P$ < .01) for temozolomide plus cisplatin compared with both high-dose IFN and observation. Grade 3 and 4 toxicities were similar between the two non-observation arms. It is described in detail in the Data Supplement (Supplement 3, Tables 2020-1 to 2020-5).

In addition to the trial by Lian et al,[113] three trials that were primarily for cutaneous melanoma reported on subgroup analyses of subsets with mucosal melanoma. Two of these trials[114,115] were found in the Cochrane review that addressed Question 3. However, as both included 16 or fewer patients with mucosal melanoma, they are not further discussed. One trial, CheckMate 238,[10] was found as part of the search to answer question 2. This trial included 27 patients with mucosal melanoma, and a subgroup analysis was not able to establish any RFS difference between ipilimumab and nivolumab (HR, 1.57; 95% CI, 0.57 to 4.33).

No new trials that met the inclusion criteria were identified in mucosal melanoma in this full update. The Panel notes that one trial reported by Yan et al[116] compared bevacizumab with no bevacizumab in patients receiving carboplatin and paclitaxel but did not meet the inclusion criteria as it had fewer than 50 patients on the control arm.

**Clinical interpretation.** While the trial by Lian et al did find improved RFS and OS, the consensus of the Expert Panel in the 2020 guideline was that the absolute median survival and RFS values seem much higher than those found in trials of cutaneous melanoma, and therefore, the results of this trial could be an artifact of its small size and design. In addition, given the results of trials in cutaneous melanoma, it is difficult to recommend chemotherapy with temozolomide and cisplatin when the indirect evidence in a similar disease would suggest that other therapies might be superior. Unlike uveal melanoma, it seems unlikely that there will ever be well-conducted trials of sufficient size and power of anti–PD-1 therapy in mucosal melanoma, and thus, this issue is unlikely to be resolved from an evidence-based perspective. Therefore, the Expert Panel felt that an expert opinion–based recommendation was appropriate in this case. As no new trials were identified in the full update, no changes to the recommendation were made. While there is no universally accepted single staging classification system for mucosal melanoma, the Panel believes that recommendations 3.1 through 3.5 can still be applied without harm being caused by potential ambiguity around exact stage equivalence.

The Expert Panel is aware of nonrandomized data[117] that suggest that KIT-positive mucosal melanoma may be responsive to imatinib. However, the consensus of the Expert Panel was that the indirect evidence in nonmucosal melanoma meant that the therapies that had been recommended in 3.1 through 3.5 were more likely to be beneficial to patients with unresectable and/or metastatic mucosal melanoma regardless of KIT status.

*Recommendation 4.3*

No recommendation for or against any specific systemic therapy for patients with any other form of noncutaneous melanoma may be made at this time. Patients should be offered or referred for enrollment in clinical trials where possible (Type: No recommendation; Evidence quality: Not applicable; Strength of recommendation: Not applicable).

**Literature review and analysis.** No randomized trials were identified for any other form of noncutaneous melanoma for either the 2020 guideline or the 2022 update.

**Clinical interpretation.** Because other forms of noncutaneous melanoma are rare, it is unlikely that there will ever be high-quality randomized data to inform the choice of therapy for these patients. Unlike the case of uveal or mucosal melanoma, however, the Expert Panel did not feel that they could recommend the evidence-based cutaneous melanoma therapies in these patients. As noted in the clinical interpretation for recommendation 3.5, testing for *NTRK* fusion and for those whose tumors have such fusions offering larotrectinib or entrectinib may be reasonable in patients with noncutaneous unresectable and/or metastatic melanoma where no other therapy is supported or who have progressed on other therapies. The prevalence of NTRK fusion in noncutaneous melanoma[101] is not well understood.

## DISCUSSION

### Ongoing Research

There are several ongoing trials that are scheduled to be completed by 2025 investigating options recommended in this guideline and that would help clarify the recommendations.

- The EBIN trial (EORTC 1612, ClinicalTrials.gov identifier: NCT03235245), a two-arm phase II trial evaluating a 3-month treatment with encorafenib plus binimetinib followed by ipilimumab (1 mg/kg once every three weeks) plus nivolumab (3 mg/kg once every three weeks) versus ipilimumab (1 mg/kg once every three weeks) plus nivolumab (3 mg/kg once every three weeks) in patients with *BRAF*-mutated stage III or IV unresectable melanoma that is fully accrued with PFS results expected in 2023.
- The STOP-GAP trial (ClinicalTrials.gov identifier: NCT02821013) of intermittent versus continuous anti–PD-1 therapy in patients with metastatic melanoma is due to be completed in 2023.
- The COWBOY trial (ClinicalTrials.gov identifier: NCT02968303) investigating induction vemurafenib plus cobimetinib followed by nivolumab plus ipilimumab versus nivolumab plus ipilimumab in patients with unresectable and/or metastatic *BRAF* V600E/K–mutated melanoma was due to be completed in 2022.
- The STARBOARD/KEYNOTE-B80 trial (ClinicalTrials.gov identifier: NCT04657991) of encorafenib plus binimetinib

versus placebo in patients with unresectable and/or metastatic *BRAF* V600E/K–mutated stage IIIB-D/IV melanoma receiving pembrolizumab is due to be completed in 2024.
- Trial NCT03978611 of continuous vemurafenib plus cobimetinib beyond progression versus second-line nivolumab or pembrolizumab in patients with unresectable and/or metastatic *BRAF* V600–mutated melanoma is due to be completed in 2023.
- The NADINA trial (ClinicalTrials.gov identifier: NCT04949113) of neoadjuvant nivolumab plus ipilimumab versus adjuvant nivolumab in patients with resectable stage III melanoma is due to be completed in 2025.
- The RELATIVITY-098 trial (ClinicalTrials.gov identifier: NCT05002569) of relatlimab plus nivolumab versus nivolumab in patients with resected stage III-IV melanoma is due to be completed in 2025.



### Anti–PD-1 Therapy Potential Equivalence

The issue of the potential equivalence, or lack thereof, of the two relevant anti–PD-1 therapies nivolumab and pembrolizumab was a source of difficulty for the Expert Panel. On the one hand, these agents have a similar mechanism of action, and across multiple trials in multiple cancers, their efficacy and toxicity have been broadly similar. On the other hand, there is no, and likely never will be, any direct head-to-head comparison of these agents in a randomized trial. As the Expert Panel worked through the available evidence, there were some cases where a sufficient consensus was present to extrapolate the evidence for one agent to another, as in the case of recommendation 2.5 and the recommendation in favor of pembrolizumab for patients with resectable stage IV melanoma. But there are other cases where this consensus was not present; for example, no serious discussion was there regarding recommending pembrolizumab plus ipilimumab or pembrolizumab plus relatlimab, which might be reasonable options if nivolumab and pembrolizumab were truly equivalent. The Expert Panel sought to be conservative in its judgments regarding these matters but recognizes that others may find that the evidence suggests more or less equivalence.

### BRAF/MEK Inhibitor Potential Equivalence

The same issues arise, although not to the same level of importance, with the three BRAF/MEK inhibitor combinations. As with the anti–PD-1 therapies, there are similar mechanisms of action for BRAF/MEK inhibitors, although different pharmacokinetics, and a very low likelihood of any future direct trial comparison. While these combinations might have different toxicity profiles and potentially vary in their benefits, there is no randomized evidence to support or refute those assertions and network meta-analyses of the available randomized data are inconclusive. Again, the Expert Panel has sought to be conservative in its recommendations and follow the trial evidence as closely as possible.

### Changes in Staging Criteria

One issue that has complicated the 2020 guideline and this full update is that many of the relevant trials were conducted before the switch between the seventh and the eighth edition AJCC staging criteria.[7] In the eighth edition, mitotic rate has been removed as the staging criterion for T1 tumors. However, mitotic rate is still an important prognostic factor that should be recorded for patients with a T1-T4 primary cutaneous melanoma. The eighth edition classifies regional node involvement as either clinically occult (found microscopically in the sentinel lymph node) or clinically detected on physical examination or by imaging. Furthermore, risk stratification by substage for stage III melanoma in the eighth edition combines both primary and nodal characteristics. This subclassification was not used as a stratification factor in any of the reported trials. Therefore, clinical benefit by contemporary substage cannot be accurately assessed. Microsatellite, in-transit cutaneous and subcutaneous metastases are formally stratified according to the number of tumor-involved lymph nodes. Patients with distant metastasis are subcategorized according to the sites of disease involvement. Staging also includes the use of serum lactate dehydrogenase level to further define each category and integration of the fourth category on the basis of the presence of central nervous metastasis.

In this guideline, the Expert Panel has attempted to use eighth edition AJCC staging criteria[7] where possible, particularly in the recommendations. However, in many of the relevant trials, the staging descriptions in the text and in the Data Supplement refer to seventh edition. The consensus of the Expert Panel is that this discrepancy, although important, does not compromise the value of the recommendations.

## PATIENT AND CLINICIAN COMMUNICATION

With all cancers, clinician expertise when informing patients about their disease, diagnosis, and treatments and when offering and recruiting patients into clinical trials is vital. Information to the patient should allow the patient to feel enabled. A patient who finds agency with the information they get is likely more motivated, more proactive, more adherent, and better able to cope with their diagnosis.

Side effect management is a crucial element of patient and clinician communication. The side effects of the recommended immunotherapies and targeted therapies will vary by patient and by agent. Clinicians should recognize that even grade 1 adverse events, if chronic, can substantially affect quality of life (eg, fatigue). Long-term management of durable side effects should be a part of patient-clinician communications, especially if it involves a transition to a family or primary care provider. ASCO has developed a guideline on immune-related events on immune checkpoint therapy,[72] as has the European Society for Medical Oncology (ESMO).[118] ASCO has also published a guideline on the screening, assessment, and management of fatigue.[119]

Patients' access to information on and opportunities to enroll in clinical trials may vary substantially depending on whether the patient is receiving care in a community versus academic center setting. Clinicians should work to inform themselves of relevant clinical trials. Clinicians may also encourage patients to seek out local, regional, and national patient support organizations. ASCO's Cancer.Net online resource provides information on such organizations in the United States.

For recommendations and strategies to optimize patient-clinician communication, see the Patient-Clinician Communication: American Society of Clinical Oncology Consensus Guideline.[120]

## HEALTH DISPARITIES

Although ASCO clinical practice guidelines represent expert recommendations on the best practices in disease management to provide the highest level of cancer care, it is important to note that many patients have limited access to medical care. Racial and ethnic disparities in health care contribute significantly to this problem in the United States. Patients with cancer who are members of racial and/or ethnic minorities suffer disproportionately from comorbidities, experience more substantial obstacles to receiving care, are more likely to be uninsured, and are at greater risk of receiving care of poor quality than other Americans.[121,122] Many other patients lack access to care because of their geographic location and distance from appropriate treatment facilities. Awareness of these disparities in access to care should be considered in the context of this clinical practice guideline, and health care providers should strive to deliver the highest level of cancer care to these vulnerable populations.

In the specific case of melanoma, there is evidence that the incidence rate of cutaneous melanoma is lower in those with dark-pigmented skin versus light-pigmented skin.[123] However, there is also evidence that African Americans are often diagnosed with melanoma at noncutaneous sites including mucosal and acral—and that trials addressing these subtypes will afford a chance to address this. Disease is also often discovered at a later stage and therefore may incur increased morbidity and mortality.[124]

## MULTIPLE CHRONIC CONDITIONS

Creating evidence-based recommendations to inform treatment of patients with additional chronic conditions, a situation in which the patient might have two or more such conditions—referred to as multiple chronic conditions—is challenging. Patients with multiple chronic conditions are a complex and heterogeneous population, making it difficult to account for all possible permutations to develop specific recommendations for care. In addition, the best available evidence for treating index conditions, such as cancer, is often from clinical trials whose study selection criteria may exclude these patients to avoid potential interaction effects or confounding of results associated with multiple chronic conditions. As a result, the reliability of outcome data from these trials may be limited, thereby creating constraints for expert groups to make recommendations for care in this heterogeneous patient population.

As many patients for whom guideline recommendations apply present with multiple chronic conditions, any treatment plan needs to consider the complexity and uncertainty created by the presence of those conditions and highlights the importance of shared decision making regarding guideline use and implementation. Therefore, in consideration of recommended care for the target index condition, clinicians should review all other chronic conditions present in the patient and take those conditions into account when formulating the treatment and follow-up plan.

## RELATED ASCO GUIDELINES

- Integration of Palliative Care Into Standard Oncology Care[125] (http://ascopubs.org/doi/10.1200/JCO.2016.70.1474)
- Patient-Clinician Communication[120] (http://ascopubs.org/doi/10.1200/JCO.2017.75.2311)
- Management of Immune-Related Adverse Events in Patients Treated With Immune Checkpoint Inhibitor Therapy[72] (https://ascopubs.org/doi/10.1200/JCO.21.01440)
- Treatment of Brain Metastases[88] (https://ascopubs.org/doi/10.1200/JCO.21.02314)
- Somatic Genomic Testing in Patients With Metastatic or Advanced Cancer[100] (https://ascopubs.org/doi/full/10.1200/JCO.21.02767)

**Related Guidelines From Other Organizations Endorsed By ASCO**

- Radiation Therapy for Brain Metastases: An ASTRO Clinical Practice Guideline[102] (https://doi.org/10.1016/j.prro.2022.02.003)

## COST IMPLICATIONS

A nonsystematic review of cost-effectiveness literature was conducted for the 2020 guideline. See that article for details and for a discussion of cost implications.

## EXTERNAL REVIEW AND OPEN COMMENT

The draft recommendations were released to the public for open comment from February 27 through March 13, 2023. Response categories of "Agree as written," "Agree with suggested modifications," and "Disagree. See comments" were captured for every proposed recommendation with many written comments received. The lowest proportion of full agreement for any recommendation was 63% of respondents, with many >75%. The highest proportion of disagreement with any recommendation was 10% of respondents. The Expert Panel considered these comments and incorporated changes into the clinical interpretation for several recommendations to clarify the language. All changes were incorporated before EBMC review and approval.

The draft was submitted to two external reviewers with content expertise. Both reviewers provided feedback on details of the guideline, but all expressed confidence in the guideline and considered it of good quality. Their comments were reviewed by the methodologist author (HM) and integrated into the final manuscript before final approval by the Expert Panel and the EBMC.

## GUIDELINE IMPLEMENTATION

ASCO guidelines are developed for implementation across health settings. Each ASCO guideline includes a member from ASCO's Practice Guideline Implementation Network (PGIN) on the Panel. The additional role of this PGIN representative on the guideline Panel is not only to assess the suitability of the recommendations to implementation in the community setting but also to identify any other barrier to implementation a reader should be aware of. Barriers to implementation include the need to increase awareness of the guideline recommendations among frontline practitioners and survivors of cancer and caregivers and also to provide adequate services in the face of limited resources. The guideline Bottom Line Box was designed to facilitate implementation of recommendations. This guideline will be distributed widely through the ASCO PGIN. ASCO guidelines are posted on the ASCO website and most often published in the *JCO*.

**ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care and that all patients should have the opportunity to participate.**

## ADDITIONAL RESOURCES

More information, including a Data Supplement with additional evidence tables, slide sets, and clinical tools and resources, is available at www.asco.org/melanoma-guidelines. Patient information is available at www.cancer.net.

## GENDER-INCLUSIVE LANGUAGE

ASCO is committed to promoting the health and well-being of individuals regardless of sexual orientation or gender identity.[126] Transgender and nonbinary people, in particular, may face multiple barriers to oncology care including stigmatization, invisibility, and exclusiveness. One way that exclusiveness or lack of accessibility may be communicated is through gendered language that makes presumptive links between sex and anatomy.[127-130] With the acknowledgment that ASCO guidelines may affect the language used in clinical and research settings, ASCO is committed to creating gender-inclusive guidelines. For this reason, guideline authors use gender-inclusive language whenever possible throughout the guidelines. In instances in which the guideline draws upon data based on gendered research (eg, studies regarding women with ovarian cancer), the guideline authors describe the characteristics and results of the research as reported.


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## EDITOR'S NOTE

This American Society of Clinical Oncology (ASCO) Clinical Practice Guideline provides recommendations, with comprehensive review and analyses of the relevant literature for each recommendation. Additional information, including a Data Supplement with additional evidence tables, slide sets, clinical tools and resources, and links to patient information at www.cancer.net, is available at www.asco.org/melanoma-guidelines.

## EQUAL CONTRIBUTION

R.S and S.A. were Expert Panel Cochairs.

## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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## AUTHOR CONTRIBUTIONS

**Conception and design:** All authors
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Members of Expert Panel Membership are listed in Appendix Table A2 (online only).



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## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

**Systemic Therapy for Melanoma: ASCO Guideline Update**

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Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians (Open Payments).

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## APPENDIX

**TABLE A1.** Recommendation Rating Definitions

| Term | Definitions |
|---|---|
| Quality of evidence | |
| High | We are very confident that the true effect lies close to that of the estimate of the effect |
| Moderate | We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different |
| Low | Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect |
| Very low | We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect |
| Strength of recommendation | |
| Strong | In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects<br>In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects<br>All or almost all informed people would make the recommended choice for or against an intervention |
| Weak | In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists<br>In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists<br>Most informed people would choose the recommended course of action, but a substantial number would not |

**TABLE A2.** Systemic Therapy for Melanoma Expert Panel Membership

| Name | Affiliation | Area of Expertise |
|---|---|---|
| Rahul Seth, DO (cochair) | SUNY Upstate Medical University, Syracuse, NY | Medical Oncology |
| Sanjiv S. Agarwala, MD (cochair) | Lewis Katz School of Medicine at Temple University, Philadelphia, PA | Medical Oncology |
| Krishna C. Alluri, MD | St Luke's Mountain States Tumor Institute, Boise, ID | Medical Oncology |
| Paolo A. Ascierto, MD | Istituto Nazionale Tumori IRCCS Fondazione Pascale, Napoli, Italy | Medical Oncology |
| Michael B. Atkins, MD | Georgetown-Lombardi Comprehensive Cancer Center, Washington, DC | Medical Oncology |
| Kathryn Bollin, MD | Scripps MD Anderson Cancer Center, San Diego, CA | Medical Oncology |
| Matias Chacon, MD | Instituto Alexander Fleming, Buenos Aires, Argentina | Medical Oncology |
| Nancy Davis, MD | Vanderbilt University Medical Center, Nashville, TN | Medical Oncology |
| Mark B. Faries, MD | The Angeles Clinic and Research Institute and Cedars Sinai Medical Center, Los Angeles, CA | Surgical Oncology |
| Pauline Funchain, MD | Cleveland Clinic, Cleveland, OH | Medical Oncology |
| Jason S. Gold, MD | VA Boston Healthcare System, West Roxbury, MA | Surgical Oncology |
| Samantha Guild, JD | AIM at Melanoma Foundation, Frisco, TX | Patient Representative |
| David Gyorki, MBBS | Peter MacCallum Cancer Centre, Melbourne, Victoria, Australia | Surgical Oncology |
| Varinder Kaur, MD | University of Virginia, Charlottesville, VA | Medical Oncology |
| Nikhil Khushalani, MD | H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL | Medical Oncology |
| John M. Kirkwood, MD | University of Pittsburgh School of Medicine and UPMC Hillman Cancer Institute, Pittsburgh, PA | Medical Oncology |
| Jennifer Leigh McQuade, MD | University of Texas MD Anderson Cancer Center, Houston, TX | Medical Oncology |
| Michael O. Meyers, MD | University of North Carolina at Chapel Hill, Chapel Hill, NC | Surgical Oncology |
| Anthony Provenzano, MD | Columbia Presbyterian Clinical Assoc, Bronxville, NY | Medical Oncology |
| Caroline Robert, MD, PhD | Gustave Roussy Cancer Center and Paris-Saclay University, Villejuif, France | Dermatology |
| Mario Santinami, MD | Fondazione IRCCS Istituto Nazionale dei Tumori, Milano, Italy | Surgical Oncology |
| Amikar Sehdev, MPH, MD | Ironwood Cancer and Research Centers, AZ | Medical Oncology |
| Vernon K. Sondak, MD | H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL | Surgical Oncology |
| Gilliosa Spurrier, MSc | Melanome France, Teilhet, France | Patient Representative |
| Umang Swami, MS, MD | Huntsman Cancer Institute, University of Utah, Salt Lake City, UT | Medical Oncology |
| Thach-Giao Truong, MD | Kaiser Permanente Northern California, Berkeley, CA | Medical Oncology |
| Katy K. Tsai, MD | Helen Diller Family Comprehensive Cancer Center, University of California, San Francisco, CA | Medical Oncology |
| Alexander van Akkooi, MD, PhD | Melanoma Institute Australia, University of Sydney and Royal Prince Alfred Hospital, Sydney, Australia | Surgical Oncology |
| Jeffrey Weber, MD, PhD | Laura and Isaac Perlmutter Cancer Center at NYU Langone Health, New York, NY | Medical Oncology |
| Hans Messersmith, MPH | American Society of Clinical Oncology, Alexandria, VA | ASCO Practice Guidelines Staff (Health Research Methods) |